Daily Briefing - November 03, 2026

Your Daily Tech & Programming Digest

Tuesday, November 03, 2026

1000	121690	1189	35
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

CAG repeat length shapes bidirectional functional connectome alterations in spinocerebellar ataxia type 3: a multimodal MRI study

📖 1 min 📝 32 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Xianfei Yang, Haishuo Xia, Yuwei Hu, Can Xu, Xingang Wang, Qingxin Deng, Qian Xie, Limeng Dai, Qifang Yang, Changying Li, Li Gui, Chen Liu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007013?dgcid=rss_sd_all

Coarse motor-state structure across paired preoperative fMRI and intraoperative ECoG in patients with brain tumors

📖 1 min 📝 36 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Jingxi Hu, Jie Ma, Wanwan Guo, Yang Jiao, Jiaqi Yang, Hongwei Wang, Yiheng Yin, Renke He, Gang Cheng, Hongmin Bai, Hairong Zheng, Dong Liang, Jianning Zhang, Zhanli Hu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006257?dgcid=rss_sd_all

Parietal structural and functional differentiation associated with visual tracking across motor participation profiles

📖 1 min 📝 26 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Keying Zhang, Meng Qi, Ling Jiang, Qing Yan, Yuyan Wang, Shuang Bai, Chunmei Cao, Dong Zhang, Siyuan Ma

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006178?dgcid=rss_sd_all

Brief Peppermint Odour Exposure and Immediate Free Recall of a Visual Object Array: A Three-Arm Randomised Experiment

👤 Akar, B., Altıntas, T.
📅 2026-11-02 📖 1 min 📝 270 words BIORXIV NEUROSCIENCE

Summary: Background: Odours reach limbic and prefrontal targets through a uniquely direct pathway and are often claimed to enhance cognition, yet experimental evidence for specific essential oils is mixed and frequently confounded by expectancy. We tested whether a brief inhalation of peppermint or clove ess...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.31.763290v1?rss=1

Myelin Governs Hippocampal Network Function through the Extracellular Space it Occupies, Not Only the Axons it Insulates

👤 Jia, X., Zeng, C., Liu, L., Gao, T., Bi, J., Zeng, Z., Liang, L., Qi, C., Lin, F., Cheng, F.
📅 2026-11-03 📖 1 min 📝 250 words BIORXIV NEUROSCIENCE

Summary: Myelin, the multilamellar sheath wrapping neuronal axons, enables saltatory conduction. Beyond electrical insulation, myelin occupies volume, and therefore shapes the extracellular space (ECS) through which interstitial fluid flows and neurotransmitter diffuses. How these two roles interact is unkno...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.02.763564v1?rss=1

Path knockout: an analysis tool for connectome-driven circuit discovery

McGrath, R., Hattori, D.
2026-11-03 1 min 207 words BIORXIV NEUROSCIENCE

Summary: Neural circuits in the brain transform sensory information into perceptual, emotional and motor output. The connectome -- a synaptic-resolution wiring diagram of all neurons in a nervous system -- provides a neuroanatomical blueprint for identifying different circuits that mediate specific computati...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.31.763128v1?rss=1

A medial entorhinal connectome reveals compartmentalized microcircuit architecture

Shi, T., Chen, Y., Jin, H., Jiang, H., Chen, X., Xiong, Z., Wu, F., Zhang, R.
2026-11-03 1 min 217 words BIORXIV NEUROSCIENCE

Summary: The medial entorhinal cortex (MEC) transforms multimodal inputs into spatial representations, but the synaptic architecture underlying its computations remains unclear. Here we generated a synapse-resolution connectomic dataset spanning all six layers of mouse MEC and the adjacent parasubiculum with...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762527v1?rss=1

Wolffia globosa -derived nanovesicles as a natural source of bioactive metabolites with neuroprotective potential against methylglyoxal-induced oxidative stress

👤 Khongkla, E., Boonsnongcheep, P., Srisaisup, M., Charuraksa, T.

📅 2026-11-03 📖 1 min 📝 219 words

BIORXIV NEUROSCIENCE

Summary: Plant-derived nanovesicles (PDNVs) have attracted increasing attention for their potential biomedical applications, particularly in oxidative stress-related diseases and neuroprotection. Wolffia globosa, an emerging functional food and potential future food source, is rich in phytochemicals associat...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.31.762905v1?rss=1

Dissociable norepinephrine and dopamine representations during strategic decision-making and reward learning

👤 Aloor, J., Sit, T. P., Geuter, J., Tutt, J. O., Duan, C. A.

📅 2026-11-03 📖 1 min 📝 256 words

BIORXIV NEUROSCIENCE

Summary: Adaptive decision-making in competitive environments requires learning from immediate reward outcomes while tracking longer-timescale information to adjust behavioural strategies and remain unpredictable against others. Dopamine (DA) and norepinephrine (NE) have long been implicated in reward-guided...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.31.763226v1?rss=1

Oscillatory Phase Coding Emerges in Adaptive RNNS for Flexible Working Memory

Lu, K., Chu, T., TUERHONG, A., Sun, Y., Luo, H., Fang, F., WU, S.

2026-11-03 1 min

210 words

BIORXIV NEUROSCIENCE

Summary: Oscillatory phase coding (OPC) is widely observed in neural systems and has been implicated in cognitive functions such as memory and sequence processing. However, its computational role and neural substrates remain largely unclear. Here, we studied OPC by training recurrent neural networks (RNNs) w...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.02.763841v1?rss=1

Multi-scale place fields provide a predictive basis for hierarchical control

Yu, C., Maimon, S. R., Rich, P. D., Eliav, T., Las, L., Lee, A. K., Ulanovsky, N., Lengyel, M.

2026-11-03 1 min

150 words

BIORXIV NEUROSCIENCE

Summary: Hippocampal place cells typically exhibit unimodal spatial selectivity in small enclosures, but express multiple, irregularly spaced fields in environments of naturalistic scale. However, existing predictive-map theories of the hippocampus do not scale well to large environments and do not account f...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762461v1?rss=1

BundleMorphometry: Spatially Resolved White Matter Bundle Shape Analysis Using Nonlinear Streamline-based Registration

Dasanayake, D., Amit, S. N., Chandio, B. Q.

2026-11-03 1 min 203 words

BIORXIV NEUROSCIENCE

Summary: Diffusion MRI tractometry quantifies microstructural properties along white matter (WM) bundles but does not directly measure their geometry, potentially overlooking localized morphological alterations. Here, we introduce BundleMorphometry, a framework for quantifying local WM bundle shape using the...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.02.763786v1?rss=1

Circuit regulation along an activity continuum shapes flexible and stable representations during learning

Yusufi, D., Ma, Y., Chen, L., Wang, G., Malone, T., Zhang, K., Kim, J. H., Shahid, F., Cekada, K., Barcelone, E., Gu, Y.

2026-11-03 1 min 207 words

BIORXIV NEUROSCIENCE

Summary: Learning requires neural representations to flexibly incorporate new information while progressively stabilizing to support enduring memories. How neural circuits balance these demands remains unclear. Here, we identify complementary mechanisms supporting representational flexibility and stability i...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762663v1?rss=1

RIPK1/RIPK3 and MLKL differentially control mitophagy and mitochondrial homeostasis in hypoxic-ischaemic encephalopathy

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41419-026-09382-w

Large-scale identification of the regulatory potential of SNPs in human neural stem cells

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-79051-1

Advancing human iPSC-derived motor neuron models using glutamatergic modulators for synaptic function studies

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41419-026-09381-x

Integrative analysis of brain-age acceleration: health associations and multi-omics profiles

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🔗 **Read full article:**
https://www.nature.com/articles/s41514-026-00557-5

Subacute carbon monoxide exposure alters AgNOR number and area ratio in the cerebral cortex of rats

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77555-w

Aging-related cellular and molecular changes reveal a role for Bglap3+ neurons in MEC spatial cognition

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https://www.nature.com/articles/s41467-026-78673-9

Early-life adversity induces social anxiety and cognitive deficits alongside neurobiological changes in rats

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🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11279-5

Orexin and pupillary modulation for hunger control

👤 Seher KosarEva BraceyNikola GrujicYihui DuDaria Peleg-RaibsteinDenis Burdakovahttps://ror.org/02q2nz798ETH Zürich, Department of Health Sciences and Technology, Neuroscience Center Zürich, Institute for Neuroscience, Schwerzenbach 8603, Switzerland

📅 2026-10-28 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThis study reveals a genetically defined arousal-hunger circuit that enables mental engagement or external sensory input to naturally interrupt the urge to eat. These findings in mice suggest that...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619759123?af=R

Dlx1&2 directly promote expression of Cxcl14 to control synapse development and interneuron survival

👤 Teng GuoSusan LindtnerJia Sheng HuSiavash Fazel DarbandiYurong WangEmily Ling-Lin PaiAthéna R. YpsilantiCindy T. J. van VelthovenAlex S. NordJohn L. R. RubensteinaNina Ireland Laboratory of Developmental Neurobiology, Department of Psychiatry and Behavioral Sciences, University of California San Francisco, San Francisco, CA 94143bDepartment of Neurobiology, Physiology, and Behavior, Center for Neuroscience, University of California Davis, Davis, CA 95618cDepartment of Psychiatry and Behavioral Sciences, Center for Neuroscience, University of California Davis, Davis, CA 95618dhttps://ror.org/03cpe7c52Allen Institute for Brain Science, Seattle, WA 98109

📅 2026-10-28 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDlxgenes control multiple steps in the development of cortical interneurons (CIN), including synapse development and cell survival. Here we explore molecular mechanisms how theDlxgenes promote neo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2616055123?af=R

Lysine crotonylation acts as an epigenetic switch for glutamate neurotransmission and spatial memory

👤 Utsav MukherjeeBudhaditya BasuPratyush SuryavanshiStacy E. BeyerSaaman GhodsiNathan RobillardYann VanrobaeysEric B. TaylorJoseph GlykysTed AbelSnehajyoti Chatterjeeahttps://ror.org/036jqmy94Department of Neuroscience and Pharmacology, Carver College of Medicine, University of Iowa, Iowa City, IA 52242bhttps://ror.org/036jqmy94Iowa Neuroscience Institute, University of Iowa, Iowa City, IA 52242chttps://ror.org/036jqmy94Interdisciplinary Graduate Program in Neuroscience, University of Iowa, Iowa City, IA 52242dhttps://ror.org/036jqmy94Stead Family Department of Pediatrics, Carver College of Medicine, University of Iowa, Iowa City, IA 52242ehttps://ror.org/036jqmy94Interdisciplinary Graduate Program in Genetics, University of Iowa, Iowa City, IA 52242fhttps://ror.org/036jqmy94Fraternal Order of Eagles Diabetes Research Center, University of Iowa, Iowa City, IA 52242ghttps://ror.org/036jqmy94Department of Molecular Physiology and Biophysics, Carver College of Medicine, University of Iowa, Iowa City, IA 52242

📅 2026-10-28 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAlthough histone acetylation has long been considered a principal regulator of gene expression events underlying memory storage, the role of non-acetyl histone acyl modifications on long-term memo...

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https://www.pnas.org/doi/abs/10.1073/pnas.2608460123?af=R

Forniceal deep brain stimulation improves memory in Rett syndrome mice by promoting the maturation of newborn hippocampal neurons

Qi WangBin TangZhenyu WuHuda Y. ZoghbiJianrong Tangahttps://ror.org/05cz92x43Jan and Dan Duncan Neurological Research Institute at Texas Children's Hospital, Houston, TX 77030bhttps://ror.org/02pttbw34Department of Pediatrics, Division of Neurology, Baylor College of Medicine, Houston, TX 77030chttps://ror.org/02pttbw34Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX 77030dhttps://ror.org/02pttbw34Program in Developmental Biology, Baylor College of Medicine, Houston, TX 77030ehttps://ror.org/02pttbw34Department of Neuroscience, Baylor College of Medicine, Houston, TX 77030fhttps://ror.org/02pttbw34HHMI, Baylor College of Medicine, Houston, TX 77030

📅 2026-10-28 📖 1 min 📝 52 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDeep brain stimulation (DBS) has long been known for its efficacy in mitigating movement disorders. More recently, it has shown promise for addressing disorders of learning and memory: for example...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2535887123?af=R

Defining and measuring proximity to criticality based on univariate time series

J. Samuel SooterAntonio J. FonteneleAndrea K. BarreiroCheng LyKeith B. HengenWoodrow L. ShewaDepartment of Physics, University of Arkansas, Fayetteville, AR 72701bhttps://ror.org/042tdr378Department of Mathematics, Southern Methodist University, Dallas, TX 75205chttps://ror.org/02nkdxk79Department of Mathematics and Applied Mathematics, Virginia Commonwealth University, Richmond, VA 23284dhttps://ror.org/01yc7t268Department of Biology, Washington University, St. Louis, MO 63130

2026-10-28 1 min 56 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceKnowledge of many complex systems comes from time series measurements of one or a few parts of the system. These dynamics depend on many detailed properties; must we keep track every detail to und...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2526984123?af=R

Computational approaches to deep learning–driven EEG analysis for neurological disease diagnosis: a systematic review

Jisu Elsa Jacob 2026-10-13 1 min 229 words FRONTIERS NEUROINFORMATICS

Summary: Electroencephalography (EEG) is a widely used non-invasive neurophysiological technique for analyzing brain dynamics and diagnosing neurological diseases. However, the non-stationary nature, low signal-to-noise ratio, and inter-subject variability of EEG signals cause significant challenges to tradi...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1778980

Short excitatory projections promote burst activity and modular functional organization in neural networks

Claus C. Hilgetag 2026-11-03 1 min 165 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Physical constraints in the brain favor short, distance-dependent connections. However, short-range projections are not merely a structural compromise; they also serve essential functional roles. Here, we explore networks with balanced excitation and inhibition, demonstrating that short excitatory p...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1965686

Persistent memory through triple-loop consolidation under stochastic unit turnover

Jianwei Lou 2026-11-03 1 min 366 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Dissipative cognitive architectures maintain computation through continuous energy expenditure, where units that exhaust their energy are stochastically replaced with fresh random state. This creates a fundamental challenge: how can persistent, context-specific memory survive when all learnable stat...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1845213

Editorial: Advances in perceptual decision making and brain oscillations

Francesco Di Gregorio 2026-11-03 1 min 0 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fncom.2026.2023605

Editorial: Machine learning integration in computational neuroscience: enhancing neural data decoding and prediction

George Giakos 2026-11-03 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fncom.2026.2030505

Exploratory evaluation of electroencephalogram-derived features for classification of mild cognitive impairment in two independent datasets

Makoto Nakane 2026-11-03 1 min 347 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: A proper diagnosis of mild cognitive impairment (MCI) is one of the important challenges in the early detection of dementia. For the classification of MCI patients and healthy controls, some electroencephalogram (EEG)-derived indices were calculated and compared. Some studies have been conducted on ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1940507

Correction: Post-stroke spasticity: advances in network mechanisms and therapeutics

Zhiqi Lin 2026-10-29 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2035182

Embodied multimodal human-computer interaction for English language learning: integrating speech, gestures, emotion, and adaptive feedback

Yao Tang 2026-10-14 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: Current English language learning systems are often limited by single-dimensional learning-state perception, insufficient modeling of affective-cognitive characteristics, and delayed or inflexible feedback regulation. To address these limitations, this study proposes an affective-cognitive gated cro...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1920739

Effectiveness of bimanual kinesthetic teaching through different controller settings

Andrej Gams 2026-10-21 1 min 94 words FRONTIERS NEUROROBOTICS

Summary: Physical human-robot interaction (pHRI) through kinesthetic teaching enables intuitive robot programming, but extending this approach to bimanual systems introduces challenges in synchronized control. Thus, the best way to control bimanual robot systems for kinesthetic teaching remains unclear. In t...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1960197

Editorial: Expanding the therapeutic dose range of cannabinoid-based therapy

Babak Baban 2026-11-03 1 min 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2028326

From beat to syntax: evolutionary and developmental paths of a shared neural resource for hierarchical processing in music and language

Tao Gong 2026-11-03 1 min 268 words FRONTIERS NEUROSCIENCE

Summary: Children who track a musical beat well tend to have stronger grammatical skills, and brief exposure to a regular rhythm can transiently improve grammatical processing. These findings raise the possibility of a shared underlying mechanism, and with it two questions: where did the link come from, and ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1967182

Vagus nerve stimulation in obesity: a systematic review and meta-analysis

Matthew A. Schiefer 2026-11-03 1 min 333 words FRONTIERS NEUROSCIENCE

Summary: BackgroundVagus nerve stimulation (VNS) has been investigated as a potential therapy for obesity because of the vagus nerve’s role in satiety, gastrointestinal signaling, autonomic regulation, and metabolic homeostasis. However, studies vary substantially in stimulation site, interface, parameters, ...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1955378

Analysis and improvement of surface electromyography-based American Sign Language hand gesture recognition

Yasuharu Koike 2026-11-03 1 min 243 words FRONTIERS NEUROSCIENCE

Summary: Surface electromyography (sEMG) allows sign language gestures to be recognized from forearm muscle activity without visual access to the hand. Gestures that produce similar forearm activation patterns, however, remain difficult to distinguish. This study examined how these similarities relate to cla...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1978950

Cybertissue neuroprosthetics: a new implantable neuroprosthetics paradigm to achieve seamless lifetime integration

Liang Guo 2026-10-27 1 min 234 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Despite decades of advances in neural interface technology, the long-term instability of implantable neuroprostheses remains a major barrier to achieving reliable lifetime performance. Chronic foreign-body reactions and fibrotic encapsulation progressively degrade the electrode-tissue int...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb442

Combined voltammetry and optical brain imaging: toward high resolution neurochemical measurements

Vladislav Zavialov, Vassiliy Tsytsarev, Anna B Volnova, Raul R Gainetdinov and Mikhail Inyushin

2026-10-27 1 min 316 words JOURNAL NEURAL ENGINEERING

Summary: The question of the mechanisms by which cellular interactions in the brain are converted into integrative behavioral phenomena is closely related to functional localization, and also to the relationship between spatial and temporal resolution when interpreting experimental data. Classical electrophy...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb32e

CORRIGENDUM: In silico development and validation of Bayesian methods for optimizing deep brain stimulation parameters (2023 J. Neural Eng. 20 036015)

Sumedh S Nagrale, Ali Yousefi, Theoden I Netoff and Alik S Widge

2026-10-29 1 min 0 words JOURNAL NEURAL ENGINEERING

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb32d

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261103064123&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261103064123&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261103064123&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261103064123&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261103064123&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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HiCP2GAN: a plug and play foundation model-based GAN for Hi-C enhancement

Oluwatosin Oluwadare 2026-11-02 1 min 65 words LOW VISION

Summary: The 3D organization of chromatin shapes gene regulation and cellular function, and Hi-C has become the primary technique for mapping chromatin interactions genome-wide. However, high-resolution data remain costly and scarce, leaving many studies with sparse contact maps that limit downstream analysi...

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Exploring Spatial Inequities and Livability: A Mixed-Methods Study Using Artificial Intelligence and Community Insights

Rodrigo Siqueira Reis 2026-11-02 1 min 62 words LOW VISION

Summary: Rapid urbanization has intensified health and social inequities through the uneven distribution of infrastructure, services, and public space. In St. Louis, Missouri (USA), these disparities remain geographically concentrated and rooted in long-standing patterns of racial segregation and disinvestme...

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Association Between Quality of Life and Visual Function in Inherited Retinal Disease: A Cross-sectional Study in a Retinal Genetics Centre

Robert E MacLaren
2026-11-02 1 min 27 words LOW VISION

Summary: CONCLUSIONS: Microperimetry is a patient-relevant measure in inherited retinal diseases. This study demonstrates to stakeholders, such as regulatory bodies, how visual function measures translate to patients' quality-of-life.

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Social Determinants of Kidney and Cardiovascular Outcomes: Prioritizing Action to Effectively Manage Cardiovascular-Kidney-Metabolic Syndrome

Tiffany M Powell-Wiley
2026-11-02 1 min 70 words LOW VISION

Summary: In 2005, the World Health Organization (WHO) created the inaugural Social Determinants of Health (SDOH) Commission to recognize how social and physical environments impact health for all populations. In doing so, they created an impetus for policy-making and governing bodies to improve the underlyin...

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Effectiveness of spectacle lenses with highly aspherical lenslets (HAL) in slowing axial elongation and spherical equivalent refraction progression in myopic and pre-myopic children: a systematic review and meta-analysis

👤 Jiaqi Yao
📅 2026-11-02 📖 1 min 📝 65 words

LOW VISION

Summary: CONCLUSION: Current evidence suggests that HAL spectacle lenses may slow axial elongation and SER progression in myopic children. In pre-myopic children, HAL lenses may offer a smaller benefit, particularly with high adherence, but the evidence is limited and of low quality, precluding a strong reco...

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Changes in self-concept and healthy behaviors following an audio instructional program on puberty among teenage girls with visual impairment: a quasi-experimental study

Ashraf Jehad Abuejheisheh

2026-11-02 1 min 51 words LOW VISION

Summary: CONCLUSION: The results of this investigation revealed that visually impaired adolescent girls had low knowledge, self-concept, and menstrual hygiene practices at puberty before the intervention. All measured outcome variables showed a significant improvement after the implementation of the audio in...

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FCM-Encoder: a novel interpretable framework with vision-language reasoning for coronary artery disease and solitary pulmonary nodule classification

👤 Nikolaos I Papandrianos
📅 2026-11-02 📖 1 min 📝 66 words LOW VISION

Summary: Although modern machine-learning models attain strong predictive performance, their clinical adoption remains limited by a lack of transparency. This concern is particularly relevant in multimodal diagnostic pipelines, where the contribution of each modality to the final prediction is more difficult...

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An integrated CT/MRI diagnostic system for pancreatic cystic lesions that supports pathological classification and risk stratification

👤 Jihui Hao
📅 2026-11-03 📖 1 min 📝 22 words LOW VISION

Summary: CONCLUSIONS: SIRS-PCL is a pathology-based, quantitative decision-support system for preoperative pathological subtype classification and pathological risk stratification of surgically evaluated neoplastic PCLs.

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Foveal Crowding and Reduced Binocular Advantage In Myopic Anisometropia Under Brief Stimulus Presentation

👤 Jianfeng Wu 📅 2026-11-03 📖 1 min 📝 32 words LOW VISION

> **Summary:** CONCLUSIONS: In optically corrected MA patients, brief stimulus presentation revealed abnormal foveal crowding in response time and a reduced binocular advantage, suggesting decreased efficiency in processing visual information under high perceptual demand.

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Blinded by inequity: a scoping review of income-related and structural disparities in eye care in the United States

👤 Carrie B Dolan 📅 2026-11-03 📖 1 min 📝 67 words LOW VISION

> **Summary:** CONCLUSION: Income-related disparities in eye care access are associated with structural and financial barriers linked to reduced access to timely services. Addressing these inequities may require greater alignment among recognized needs, promising interventions, and policy priorities, as well as en...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

> **Summary:** This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Effect of electrical stimulation on remyelination in preclinical models of multiple sclerosis: A systematic review

👤 Hamdan Hamdan 📅 2026-10-30 📖 1 min 📝 33 words TDCS TACS TRNS

> **Summary:** CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Dosage and safety of non-invasive brain stimulation for chronic musculoskeletal pain management: a scoping review

👤 Liming Jiang
📅 2026-10-30 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: CONCLUSION: Current NIBS dosage for management of CMP remains heterogeneous and driven by empirical experience. Safety reporting is inadequate. Although reported adverse events were predominantly mild and transient, inconsistent adverse event ascertainment and reporting preclude reliable conclusion ...

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

Qianxiang Zhou 2026-11-02 1 min 57 words TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

👤 Roi Cohen Kadosh 📅 2026-11-02 📖 1 min 📝 48 words TDCS TACS TRNS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

👤 Tobias Grossmann 📅 2026-10-29 📖 1 min 📝 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

👤 Chen-Hui Pei
📅 2026-10-30 📖 1 min 📝 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

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EEG feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

Aishi Zhou 2026-10-30 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

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A research on the typical images of animals in the dreams of Naxi.

📅 2026-09-10 📖 1 min 📝 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna’s perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000349

When the body suffers, the mind dreams: The lived experience of cancer.

📅 2026-09-10 📖 1 min 📝 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 571 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/mar-2026

Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 954 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/june-2026

Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 700 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/aug-2026

Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1079 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/sept-2026

What is the closest equivalent to AstroJS in the Python ecosystem?

👤 /u/sentientanus69 📅 2026-11-02 📖 1 min 📝 75 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I am aware a full blown clone wouldn't be possible (hard to imagine how islands would work with Python). But the part where Astro is a free form static site framework is pretty neat and I don't think the established players in this space within Python world are as m...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xmcpk7/what_is_the_closest_equivalent_to_astrojs_in_the/

Open-Source 5G Ran Platforms: A Dual Perspective on Performance and Capabilities

👤 teleforce 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://arxiv.org/abs/2608.25820 Comments URL: https://news.ycombinator.com/item?id=50347464 Points: 39 # Comments: 4

🔗 **Read full article:**
https://arxiv.org/abs/2608.25820

Show HN: I built a little browser game to practice Vim commands

👤 dh1011 📅 2026-11-03 📖 1 min 📝 62 words HACKER NEWS

> **Summary:** I made hjkl to help myself remember Vim commands. It’s a little browser game you can play when you want to unwind and get a bit of Vim practice in. There are still a lot of commands I want to add, but I hope someone finds it useful in the meantime :) <hr /> Comments URL: <a href="https:/...

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https://hjkl.duy.run

Nvidia – open-source auto-ontology

👤 campervans 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/NVIDIA/auto-ontology Comments URL: https://news.ycombinator.com/item?id=50348962 Points: 91 # Comments: 30

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https://github.com/NVIDIA/auto-ontology

uBlock Origin filter to block autoplay videos in TechCrunch

👤 cornfieldlabs 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://cornfieldlabs.github.io/posts/ublock-stop-techcrunch-autoplay-video/post.html Comments URL: <a href="https://news.ycombinator.com/item?id=50349155">https://news.ycombina...

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https://cornfieldlabs.github.io/posts/ublock-stop-techcrunch-autoplay-video/post.html

Revolut moves to self-hosted GLM

👤 pretext 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://twitter.com/guilleflorvs/status/2117379073785696395 Comments URL: https://news.ycombinator.com/item?id=50350004 Points: 87 <...

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Easily download files from the Open Science Framework with Papercheck

noreply@blogger.com (Daniel Lakens)
2025-07-22 3 min 765 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/easily-download-files-from-open-science.html

How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11 21 min 4292 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/how-i-use-ai-in-my-work.html

Visuospatial attention modulates audiovisual temporal perception: Context-dependent associations between prestimulus alpha lateralization and perceptual reports

📖 1 min 📝 30 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Zeliang Jiang, Yuqing Zhao, Weicong Ren, Jie Zhang, Xinyuan Liu, Fei Yu, Xingwei An, Wenxiao Zhong, Di Wu, Pan Zhang, Zhijie Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600635X?dgcid=rss_sd_all

The visual cortex and beyond: investigating long-term cortical plasticity following monocular deprivation

📖 1 min 📝 29 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Franziska Wagner, Lukas Grajewski, Alexander Schmidt, Ulrike Teschner, Markus Ullsperger, Kathrin Finke, Christian Gaser, Robert Dahnke, Carsten M. Klingner, Stefan Brodoehl

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600604X?dgcid=rss_sd_all

Distinct neural patterns of valence and arousal in the developing brain: A representational similarity analysis of MEG

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Ioannis Ntoumanis, Christos Papadelis

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006324?dgcid=rss_sd_all

Single-molecule bouncing at a liquid interface governs presynaptic enrichment

👤 Kudo, K., Longfield, S. F., McCann, A. J., Schoovaerts, N., Sanchez-Velasquez, J., Parra-Rivas, L. A., Roy, S., Verstreken, P., Hinde, E., Millard, S. S., Bademosi, A. T., Meunier, F. A.

📅 2026-11-02 📖 1 min 📝 187 words BIORXIV NEUROSCIENCE

> **Summary:** Many cellular processes rely on accumulating highly diffusive cytoplasmic proteins to a specific location within the cell. This spatial enrichment is generally thought to rely on locally immobilizing proteins or slowing diffusion through binding and molecular crowding. Paradoxically, cytoplasmic pro...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.01.763503v1?rss=1

Prolactin Receptor Long Isoform ΔS1 Expression is Reduced in the Dorsal Root Ganglia of Humans with Chronic Pain

He, L., Natarajan, K., Shiers, S. I., Arendt-Tranholm, A., Sayani, R., Jagan, S., Hossain, A., Rosenfield, J., Rodriguez Concepcion, A. J., Dussor, G., Akopian, A. N., Price, T. J.

2026-11-02 1 min 301 words BIORXIV NEUROSCIENCE

Summary: Prolactin receptor (PRLR) isoforms, specifically the short (PRLR-S) and long (PRLR-L), are differentially expressed in pathological conditions and may be involved in the induction and maintenance of pain in females. PRLR sex-dependent actions are well studied in rodent models of chronic pain, but th...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.27.762326v1?rss=1

EigenBrain maps cortical molecular architecture from functional, geometric, and structural eigenmodes

Wan, B., Du, P., Ferrari, A., Saberi, A., John, A., Yan, J., Liu, X., Hansen, J. Y., Liu, Z.-Q., Hong, S.-J., Carruzzo, F., Kaliuzhna, M., Geffen, T., Katthagen, T., Misic, B., Thompson, P. M., Van De Ville, D., Margulies, D. S., Warrier, V., Bethlehem, R. A. I., Schlagenhauf, F., Kaiser, S., Bernhardt, B., Valk, S. L., Kirschner, M.

2026-11-02 1 min 232 words BIORXIV NEUROSCIENCE

Summary: Neurotransmitter receptor and transporter distributions shape neural communication across the cortex and underpin mechanistic interpretations of brain function. Magnetic resonance imaging (MRI)-derived eigenmodes, low-dimensional spatial patterns that capture the principal axes of cortical organizat...

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https://www.biorxiv.org/content/10.64898/2026.10.27.758642v1?rss=1

Bone transport induces CGRP release from sensory nerves to promote angiogenesis and heal diabetic foot wounds in rats

2026-11-03 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s41598-026-73891-z

Proapoptotic RHG proteins and mitochondria are required for sensory dendrite pruning in Drosophila

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-77081-9

An exploratory remote pilot study of environmental risk factors for Lewy body dementia considering sex differences

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41531-026-01603-5

Safety of lecanemab in real-world patients with Alzheimer’s disease: a prospective, observational case series supported by FAERS signal analysis

Erhao Bao 2026-10-02 1 min 354 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveLecanemab, a humanized immunoglobulin monoclonal antibody, has been approved for the treatment of Alzheimer’s disease (AD). Although clinical trials have demonstrated its favorable safety profile, real-world studies are crucial for identifying rare adverse events (AEs) and evaluating post-m...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1977226

Correction: The PREP pipeline: standardized preprocessing for large-scale EEG analysis

Kay A. Robbins 2026-10-12 1 min 0 words FRONTIERS NEUROINFORMATICS

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.2015651

EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

Han Jong Jun 2026-10-20 1 min 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1971441

Are cortical microcircuits optimized for information flux?—A simulation-based reverse engineering study

Patrick Krauss 2026-10-20 1 min 271 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionA sufficiently large information flux in recurrent neural networks, quantified by the mutual information between successive network states, has been proposed as a marker of rich information-processing capabilities. This raises the question of whether biological neural networks, such as c...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1886601

Efficient neuroanalytic local–global learning with dual-branch CNN–lightweight Swin Transformer for Alzheimer’s disease analysis

Arshad Ali 2026-10-29 1 min 287 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Alzheimer’s disease (AD) is one of the main causes of dementia, where early detection is critical for postponing the progression of the disease and timely treatment. Recent computer-aided diagnosis using deep learning with structural MRI has shown optimal performance; however, the rising number of m...

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Virtual brain models: a roadmap towards effective brain digital twins

Viktor Jirsa 2026-10-29 1 min 205 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Virtual Brain Models (VBMs) are emerging computational tools promising to revolutionize personalised and precision medicine. VBMs reproduce brain functional dynamics through a generative model that can be constructed and optimized against single subject data, thereby generating what is called a brai...

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A unified fixed-wavelet transformer decodes eight EEG classes across motor imagery, P300, and SSVEP paradigms

Hong-Gi Yeom 2026-10-29 1 min 314 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionMotor imagery (MI) can support self-initiated control, whereas brain–computer interfaces (BCIs) based on the P300 or the steady-state visual evoked potential (SSVEP) rely on ongoing visual stimulation. Repetitive visual stimulation can cause visual fatigue; when the stimuli are suspended...

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Resolving the apparent paradox of multisensory expertise: coupling priors, common-cause priors, and how to distinguish among them

Pier Luigi Sacco 2026-10-29 1 min 348 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Perceptual expertise is widely assumed to refine multisensory perception, yet the empirical literature appears inconsistent. Trained musicians show narrower audiovisual temporal integration windows and quicker segregation, whereas experienced wine tasters seem more influenced by a wine's color in th...

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Editorial: Emerging talents in human neuroscience: psychophysiology and cognitive neuroscience 2025

Gaia Risso 2026-10-29 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2016840

Simulation-based neural dynamics of stress-related negative cognitive bias in depressive symptoms: a dynamical EEG/ERP framework

Jianping Liu 2026-10-29 1 min 192 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionThis simulation-based computational study develops a non-linear state-space framework for stress-related negative cognitive bias and multivariate EEG/ERP observations.MethodsA three-state model represents negative bias, cognitive control, and arousal, while symptom load and normalized mo...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1972914

Effect of robotic assistance on upper-limb sensorimotor recovery: a systematic review and meta-analysis

Sivakumar Balasubramanian 2026-10-20 1 min 359 words FRONTIERS NEUROROBOTICS

Summary: IntroductionRobot-assisted upper-limb neurorehabilitation enables several beneficial therapy components, including repetitive task-specific practice, high training intensity, real-time feedback, gamification, and continuous physical interaction with the patient’s limb. However, conventional comparis...

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Dolphin-inspired adaptive coordination for online multi-robot path planning in dynamic environments

Lijuan Chen 2026-10-29 1 min 200 words FRONTIERS NEUROROBOTICS

Summary: IntroductionOnline multi-robot path planning requires repeated joint motion decisions amid static obstacles, moving obstacles, and other robots.MethodsThis study introduces the Dolphin Cooperative Foraging Optimizer (DCFO), a population-based method that adapts its search strategy within each replan...

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From confidence to caution: conformal gating for cross-session motor imagery brain–computer interfaces

Sixue Xing and Aarthy Nagarajan
2026-10-22 1 min 278 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Erroneous motor imagery brain–computer interface (MI-BCI) commands can cause unintended device actions, yet standard classifiers provide no mechanism to withhold uncertain predictions. We evaluate whether conformal prediction (CP) can serve as a command gate under cross-session EEG shift,...

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https://iopscience.iop.org/article/10.1088/1741-2552/aead26

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt
2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

> **Summary:** Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

> **Summary:** Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

Suzy J Styles 2025-12-08 1 min 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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All-optical 2D perovskite photodetector arrays for pixelated long-wave infrared imaging

Peijun Guo 2026-11-01 1 min 55 words LOW VISION

Summary: Infrared detection and imaging are important for applications including thermal monitoring, night vision, environmental sensing, and molecular identification. All-optical mid-wave infrared (MWIR) and long-wave infrared (LWIR) detection offers a compelling alternative to conventional infrared detecto...

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Inequalities in the distribution of the family physician workforce in the kingdom of Saudi Arabia: a regional analysis

👤 Mohammed Khaled Al-Hanawi 📅 2026-11-01 📖 1 min 📝 46 words LOW VISION

Summary: CONCLUSIONS: Although the national distribution of family physicians appears balanced, Lorenz curve analysis reveals significant inequalities across regions, sectors, and demographic groups. These findings have implications for workforce planning and monitoring, particularly in regions exhibiting co...

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Assessing effects of low-dose CT attenuation correction on quantification and image quality on a LAFOV PET system with a modified NEMA-IQ phantom

👤 Paul K Marsden 📅 2026-11-02 📖 1 min 📝 77 words LOW VISION

Summary: CONCLUSIONS: Even for low-dose CT protocols, PET quantification remained consistent in all NEMA-IQ spheres for many materials. Beam hardening and photon starvation in the CT image are likely to occur in specific body regions such as hips, shoulders or the abdomen for an arms-down scan and these effe...

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Functional and gain-based vestibulo-ocular measures across serum 25(OH)D categories: a dissociation between fHIT performance and vHIT gain

👤 Alper Köycü 📅 2026-11-02 📖 1 min 📝 30 words LOW VISION

Summary: CONCLUSIONS AND SIGNIFICANCE: Low 25(OH)D was accompanied by reduced functional vestibulo-ocular performance in the vertical canal planes despite normal gains. fHIT may capture subclinical change that vHIT gain does not.

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HiCP2GAN: a plug and play foundation model-based GAN for Hi-C enhancement

👤 Oluwatosin Oluwadare 📅 2026-11-02 📖 1 min 📝 65 words LOW VISION

Summary: The 3D organization of chromatin shapes gene regulation and cellular function, and Hi-C has become the primary technique for mapping chromatin interactions genome-wide. However, high-resolution data remain costly and scarce, leaving many studies with sparse contact maps that limit downstream analysi...

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Exploring Spatial Inequities and Livability: A Mixed-Methods Study Using Artificial Intelligence and Community Insights

👤 Rodrigo Siqueira Reis 📅 2026-11-02 📖 1 min 📝 62 words LOW VISION

Summary: Rapid urbanization has intensified health and social inequities through the uneven distribution of infrastructure, services, and public space. In St. Louis, Missouri (USA), these disparities remain geographically concentrated and rooted in long-standing patterns of racial segregation and disinvestme...

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Association Between Quality of Life and Visual Function in Inherited Retinal Disease: A Cross-sectional Study in a Retinal Genetics Centre

👤 Robert E MacLaren 📅 2026-11-02 📖 1 min 📝 27 words LOW VISION

Summary: CONCLUSIONS: Microperimetry is a patient-relevant measure in inherited retinal diseases. This study demonstrates to stakeholders, such as regulatory bodies, how visual function measures translate to patients' quality-of-life.

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Social Determinants of Kidney and Cardiovascular Outcomes: Prioritizing Action to Effectively Manage Cardiovascular-Kidney-Metabolic Syndrome

👤 Tiffany M Powell-Wiley 📅 2026-11-02 📖 1 min 📝 70 words LOW VISION

Summary: In 2005, the World Health Organization (WHO) created the inaugural Social Determinants of Health (SDOH) Commission to recognize how social and physical environments impact health for all populations. In doing so, they created an impetus for policy-making and governing bodies to improve the underlyin...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

👤 Alan J Emanuel
📅 2026-10-25 📖 1 min 📝 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

Felix Blankenburg 2026-10-30 1 min 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

Hamdan Hamdan 2026-10-30 1 min 33 words TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Dosage and safety of non-invasive brain stimulation for chronic musculoskeletal pain management: a scoping review

👤 Liming Jiang
📅 2026-10-30 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: CONCLUSION: Current NIBS dosage for management of CMP remains heterogeneous and driven by empirical experience. Safety reporting is inadequate. Although reported adverse events were predominantly mild and transient, inconsistent adverse event ascertainment and reporting preclude reliable conclusion ...

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

Chuhuai Wang
2026-10-31 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

👤 Taomei Guo 📅 2026-10-31 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

👤 Isao Matsushita 📅 2026-10-31 📖 1 min 📝 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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https://pubmed.ncbi.nlm.nih.gov/42959558/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261102223750&v=2.20.1

Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42962872/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261102223750&v=2.20.1

Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42964949/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261102223750&v=2.20.1

Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42966412/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261102223750&v=2.20.1

Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42962808/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42962828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42963327/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42964054/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words

BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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https://pubmed.ncbi.nlm.nih.gov/42964133/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42964381/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42966015/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42966748/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

Coping strategies and dreams—The impact of waking-life distress on negative dream emotions.

2026-05-18 1 min 183 words DREAMING

Summary: The relationship between waking life and dreaming seems complex, and various studies show that dreams are influenced by waking-life experiences. However, only a few studies deal with the interactions between coping strategies in waking life and dream emotions. The main goal of this study is to inves...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000346

Tuesday Daily Thread: Advanced questions

👤 /u/ AutoModerator 📅 2026-11-03 📖 1 min 📝 288 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Wednesday Thread: Advanced Questions 🐍</h1> Dive deep into Python with our Advanced Questions thread! This space is reserved for questions about more advanced Python topics, frameworks, and best practices. <h2>How it Works:</h2> Ask A...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xmdsr0/tuesday_daily_thread_advanced_questions/

🏔️ Garhwali Language Library v1.3.0 is now live on PyPI!

👤 /u/ National_Ad5713 📅 2026-11-03 📖 1 min 📝 150 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> An open-source Python toolkit bringing Garhwali language, linguistics and cultural heritage into the digital ecosystem. Key highlights: • 2.39M+ morphological word forms
 • 3,619+ lexical lemmas across 27 domains
 • 5 Garhwali dialects
 ...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xmg1bw/garhwali_language_library_v130_is_now_live_on_pypi/

Pip 26.3 - Extended Pylock Support, Build Backend Warnings, and a caution on wheel installation

👤 /u/zurtex 📅 2026-11-03 📖 3 min 📝 614 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Today I got to be the release manager for pip 26.3! There are two big features, and one documentation update, that I think worth talking about: Extended Pylock Support You can now install extras and non-default groups from a pylock fil...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xmgv01/pip_263_extended_pylock_support_build_backend/

SLAC researchers find proof of altermagnetism, a new magnetic phase

👤 gmays 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www6.slac.stanford.edu/news/2026-10-29-slac-researchers-find-proof-altermagnetism-new-magnetic-phase Comments URL: <a href="https://news.ycombinat...

🔗 **Read full article:**
https://www6.slac.stanford.edu/news/2026-10-29-slac-researchers-find-proof-altermagnetism-new-magnetic-phase

V100 SXM2 pinout and NVLink reference

👤 airhangerf15 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/pigeon-labs-stack/v100-sxm2-reference Comments URL: https://news.ycombinator.com/item?id=50344912 Points: 13 # Co...

🔗 **Read full article:**
https://github.com/pigeon-labs-stack/v100-sxm2-reference

The SEC Should Block Anthropic's IPO to Protect the Public

👤 cdrnsf 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.techpolicy.press/the-sec-should-block-anthropics-ipo-to-protect-the-public/ Comments URL: <a href="https://news.ycombinator.com/item?id=50344919">https://news.ycom...

🔗 **Read full article:**
https://www.techpolicy.press/the-sec-should-block-anthropics-ipo-to-protect-the-public/

WorkOS reports security breach

👤 werrett 📅 2026-11-02 📖 1 min 📝 217 words HACKER NEWS

> **Summary:** It's not on their website or Trust center yet, but we received the following breach notification from WorkOS: We are writing to notify you of a data security incident that we detected on October 30, 2026 and are continuing to investigate. We value your trust and want to keep you informed while ...

🔗 **Read full article:**
https://news.ycombinator.com/item?id=50345712

GUI64 – GUI for the Commodore 64 running on top of the built-in ROM

👤 ethanpil 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.c64-wiki.com/wiki/GUI64 Comments URL: https://news.ycombinator.com/item?id=50345735 Points: 12 # Comments: 1

🔗 **Read full article:**
https://www.c64-wiki.com/wiki/GUI64

Scaling Git at OpenAI

👤 makeramen 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://openai.com/index/scaling-git-at-openai/ Comments URL: https://news.ycombinator.com/item?id=50345924 Points: 56 # Comments: 11

🔗 **Read full article:**
https://openai.com/index/scaling-git-at-openai/

Intel stock collapse worsens as $3.5T chip rout spreads

👤 entryleveljobsc 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: <a href="https://finance.yahoo.com/markets/article/the-intel-stock-collapse-is-getting-worse--and-the-37-trillion-chip-rout-is-spreading-212154947.html">https://finance.yahoo.com/markets/article/the-intel-stock-collapse-is-getting-worse--and-the-37-trillion-chip-rout-is-spreading-212...

🔗 **Read full article:**
https://finance.yahoo.com/markets/article/the-intel-stock-collapse-is-getting-worse--and-the-37-trillion-chip-rout-is-spreading-212154947.html

Ask HN: Does anyone have a contact with Cleveland-area Ham club?

👤 stockresearcher 📅 2026-11-03 📖 1 min 📝 80 words HACKER NEWS

> **Summary:** I filled out the web form for the Cuyahoga Amateur Radio Club but it's probably been circular-filed... I gave some info about a large collection of radio gear that needs to be removed from a home or it will be put into a dumpster. A lot of it brand-new. If you know anybody at the club, please ...

🔗 **Read full article:**
https://news.ycombinator.com/item?id=50346164

DiscImageStudio: burn visible artwork onto CDs, DVDs and Blu-ray discs

👤 bpierre 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/JiaFeiMiao-K-Cat/DiscImageStudio Comments URL: https://news.ycombinator.com/item?id=50346849 Points: 59 # Comments: 16...

🔗 **Read full article:**
https://github.com/JiaFeiMiao-K-Cat/DiscImageStudio

Show HN: Hacker Times, a New York Times-Inspired Version Hacker News

👤 polpenn 📅 2026-11-03 📖 1 min 📝 33 words HACKER NEWS

> **Summary:** I thought it would be fun to browse HN as if it were a newspaper. The site updates once an hour. Enjoy! <hr /> Comments URL: https://news.ycombinator.com/item?id=50346981 Points: 61 # Comments: 19

🔗 **Read full article:**
https://hackertimes.ca/

Dogmatic Bayesianism Disorder

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-12-06 📖 4 min 📝 910 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

noreply@blogger.com (Daniel Lakens)
2026-02-08 14 min 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

Steve Luck
2024-02-04 2 min 569 words

ERP BOOT CAMP

Summary: <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

Adriel Carridice 2025-11-04 1 min 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Disrupted imagery, disrupted scenes: mental scene construction in behavioural-variant frontotemporal dementia

1 min 18 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Mohamad El Haj, Guillaume Chapelet, Dimitrios Kapogiannis, Claire Boutoleau-Bretonnière

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S000689932600483X?dgcid=rss_sd_all

Impact of hearing loss on visual selective attention: Evidence from N2pc and resting-state networks

📖 1 min 📝 28 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Tongao Zeng, Xiaoou Bu, Han Zhu, Ziyao Liu, Hongli Xiao, Fuxiang Li, Wenjun Zhang, Fuyi Yang, Jie Chen, Hailin Ma

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006361?dgcid=rss_sd_all

Resource competition controls the length-dependence of synaptic protein synthesis

👤 Rubisch, P., Raak, S. B., Wodke, J. A. H., Stefan, M. I., O'Donnell, C. 📅 2026-11-02 📖 1 min

📝 140 words BIORXIV NEUROSCIENCE

> **Summary:** Synaptic plasticity is dependent on the availability of specific proteins at the synapse with distinct requirements for potentiation and depression. Specifically, experimental evidence suggests an upregulation of short proteins and a downregulation for long proteins during long-term depression (LTD)...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763010v1?rss=1

In-vivo MRI of noradrenergic and dopaminergic nuclei predicts future cognition and Alzheimers disease progression

Zhao, W., Guzman, S. J., Yoo, H. J., Ruiz, K., Chen, K., Moheban, A., Nation, D. A., Dahl, M. J., Mather, M.

2026-11-02 1 min 162 words BIORXIV NEUROSCIENCE

Summary: Noradrenergic and dopaminergic nuclei modulate cognition, and post-mortem data indicate they are affected early in the neurodegenerative cascade of Alzheimers disease. We derived in-vivo locus coeruleus (LC) and substantia nigra-ventral tegmental area (SN-VTA) MRI integrity measures from 739 ADNI pa...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.29.762942v1?rss=1

Assessing reinforcement learning and vigor with an online decision-making battery

Wenke, S., Granwald, T., Ferrante, A., Nilsson, M., Ekström, I., Papenberg, G., Guitart-Masip, M.

2026-11-02 1 min 191 words BIORXIV NEUROSCIENCE

Summary: Reinforcement learning and vigor are two fundamental aspects of decision-making modulated by dopamine. Although vigor is highly relevant for mood disorders, it has been overlooked by previous batteries characterizing decision-making abilities. We developed a test battery with two RL tasks (RL/workin...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763028v1?rss=1

Natural predator-associated chemosignals evoke distinct spatiotemporal patterns of risk assessment and vomeronasal population responses

Wang, J., Haran, V., Meeks, J. P.

2026-11-02 1 min 208 words BIORXIV NEUROSCIENCE

Summary: Natural predator-associated chemosignals elicit diverse defensive behaviors, yet it remains unclear how complex natural cues shape the spatiotemporal organization of risk assessment and activate vomeronasal sensory neurons (VSNs) and downstream circuits in the mammalian accessory olfactory system. H...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763133v1?rss=1

Sex differences in fear extinction are conditional, not categorical

Nguyen, M. T., Didachos, A., Roozendaal, B., Henckens, M. J. A. G., Gulmez Karaca, K.

2026-11-02 1 min 233 words BIORXIV NEUROSCIENCE

Summary: Threat memories are dynamic, and their behavioral expression can change with subsequent experience and the passage of time. Extinction training can reduce conditioned fear responses, yet these responses may re-emerge over time through spontaneous recovery. Although biological sex has been implicated...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.27.762318v1?rss=1

Feedback reliability modulates learning-related dynamics of midfrontal error monitoring

Gawlowska, M.

2026-11-02 1 min 153 words BIORXIV NEUROSCIENCE

Summary: Learning from feedback requires the brain to evaluate both performance outcomes and the reliability of the information used for learning. This study examined the dynamics of neural performance monitoring during deterministic (100% reliable) and probabilistic (75% reliable) associative learning using...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.30.762838v1?rss=1

Endothelin-1 from Tongue Glands Gates 1 Sodium Taste and Intake

Chen, L.-T., Xia, X., Soto, E. R., Ostroff, G. R., Zhang, J.

2026-11-02 1 min 181 words

BIORXIV NEUROSCIENCE

Summary: Sodium is an essential nutrient. To survive sodium deficiency, animals dramatically increase their appetite for salt. However, whether and how sodium depletion impacts the taste of salt to drive sodium intake is unknown. Here we show that tongue gland-derived endothelin-1 (EDN1) regulates the amilor...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762212v1?rss=1

Mapping Endogenous Psychedelic Chemistry: Indolethylamine N-Methyltransferase (INMT) Localizes Proximal to the Golgi Apparatus of Human Neurons

Will, K., Lazaro Martinez, S., Reis, S. A., Haggarty, S. J.

2026-11-02 1 min 211 words

BIORXIV NEUROSCIENCE

Summary: N,N-dimethyltryptamine (DMT) is a serotonergic psychedelic with a history of ethnobotanical use and emerging clinical efficacy for neuropsychiatric disorders. Following Julius Axelrod's pioneering studies showing DMT production in human tissues, DMT has been proposed as an endogenous neuromodulator ...

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LAMA2-related muscular dystrophy alters Schwann cell calcium homeostasis and increases peripheral nerve inflammation

👤 Wang, J., Khati, M., Heron, M., Munir, A., Akinyele, O., Johnson, M. A., Vockley, J., Pandey, U. B., Kemaladewi, D. U.

📅 2026-11-02 📖 1 min 📝 171 words BIORXIV NEUROSCIENCE

Summary: LAMA2-related muscular dystrophy (LAMA2-RD) affects skeletal muscle and peripheral nerves, yet the consequences of LAMA2 deficiency on peripheral nerve integrity beyond early postnatal development remain incompletely understood. We investigated sciatic nerve architecture and molecular alterations ac...

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https://www.biorxiv.org/content/10.64898/2026.10.27.762325v1?rss=1

Nucleus accumbens acetylcholine dips track reward value and temporal costs to shape delay discounting

Cavallaro, J., Zuelke, D. R., Maysonet, C. M., Salazar, R. E., Setara, R., Gallo, E. F.

2026-11-02

1 min

186 words

BIORXIV NEUROSCIENCE

Summary: As an organism chooses between competing rewards, it must often integrate learned reward value with the changing costs associated with obtaining them. Yet, the brain mechanisms through which these factors jointly guide choice are not well understood. Previous studies have shown that dopamine signals...

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https://www.biorxiv.org/content/10.64898/2026.10.30.760841v1?rss=1

Anterior cingulate neurons perform parallel processing of fixated and peripheral information to guide information sampling and choice

2026-11-02

1 min

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Human subthalamic neurons track speech features and couple with auditory cortex during preparatory listening

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iTBS and cTBS to an individualized dlPFC target produce differential working memory improvement and session-dependent putamen deactivation

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A brain of many microglia origins

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Misfolded tau propagates via templated seeding

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Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

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Sleep-sensitive neurons mediate consolidation

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It’s time to split

👤 William P. Olson 📅 2026-11-02 📖 1 min 📝 12 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02511-w It’s time to split

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https://www.nature.com/articles/s41593-026-02511-w

A brain of many microglia origins

Ioana A. Marin 2026-11-02 1 min 14 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02509-4 A brain of many microglia origins

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Correcting automatic decisions

Luis A. Mejia 2026-11-02 1 min 11 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02510-x Correcting automatic decisions

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Misfolded tau propagates via templated seeding

Qingzhong Ren 2026-11-02 1 min 14 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02512-9 Misfolded tau propagates via templated seeding

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Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

Michael W. Cole 2026-11-02 1 min 22 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02530-7 Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

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From pose estimation accuracy to motor biomarker reliability: a reliability-centred evaluation framework for markerless movement analysis

Christine Fernandez-Maloigne
2026-10-01 1 min 246 words

FRONTIERS NEUROINFORMATICS

Summary: Markerless pose estimation has become an essential tool for quantitative movement analysis in neuroscience. However, current evaluation frameworks primarily rely on frame-wise spatial accuracy metrics, which provide only limited information about the reliability of the downstream kinematic biomarker...

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Altered interictal alpha-band organization in childhood absence epilepsy

Justyna Paprocka
2026-10-23 1 min 318 words

FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

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Short-term synaptic plasticity modulates the outcome of neurodegenerative diseases

Chi Chung Alan Fung 2026-10-23 1 min 264 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAlzheimer's disease (AD) is characterized by progressive neuronal loss and synaptic dysfunction that disrupt memory-encoding neural circuits. Short-term synaptic plasticity (STP), which governs transient activity-dependent changes in synaptic efficacy on millisecond-to-second timescales,...

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Orientation detection for self-calibrating neuromorphic systems

Elisabetta Chicca 2026-10-23 1 min 183 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Bioinspired sensory architectures help build adaptive touch systems with efficient hardware implementations. Orientation detection is a fundamental capability for manipulating objects in real-world scenarios, with diverse applications in robotic manipulation and manufacturing assessment processes. I...

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A systems-level energetic framework for conscious processing: energetic organization as a constraint on large-scale neural integration

Holly Pollard-Wright 2026-10-23 1 min 259 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Conscious processing is associated with coordinated interactions among distributed neural systems, yet the biological constraints that enable and sustain large-scale neural integration remain incompletely understood. Contemporary theories emphasize computational, informational, and network mechanism...

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Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

Yasushi Takagi 2026-10-23 1 min 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Early applications of apple vision pro in clinical medicine and medical education: a scoping review

👤 Chuxu Wang 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: The Apple Vision Pro (AVP), a next-generation spatial computing device, has demonstrated potential applications in multidisciplinary clinical diagnosis, treatment, and medical education, by leveraging its unique technical architecture and performance advantages. In clinical practice, its fundamental...

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A Distributed Vision Transformer-Based Deep Learning Framework for Early Detection and Classification of Lumbar Disc Degeneration

👤 N Chaitanya Kumar 📅 2026-10-31 📖 1 min 📝 68 words LOW VISION

Summary: Lumbar disc degeneration (LDD) is a principal structural cause of chronic low back pain. Magnetic resonance imaging (MRI) is the reference modality for assessing the intervertebral discs, but severity grading is performed manually, is limited by inter-observer variability, and becomes a bottleneck a...

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A Result-Auditable Protocol for Low-Data Classification of Blue-and-White Porcelain Botanical Motifs Using Frozen Chinese-CLIP Features

👤 Chenghao Yu
📅 2026-10-31 📖 1 min 📝 58 words

LOW VISION

Summary: Low-data visual classification is common in cultural heritage documentation, where expert labels are expensive, and image sources are heterogeneous. This protocol provides result-level audit support for classifying five botanical motifs-peony, lotus, chrysanthemum, plum blossom, and bamboo-on blue-a...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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All-optical 2D perovskite photodetector arrays for pixelated long-wave infrared imaging

Peijun Guo 2026-11-01 1 min 55 words LOW VISION

Summary: Infrared detection and imaging are important for applications including thermal monitoring, night vision, environmental sensing, and molecular identification. All-optical mid-wave infrared (MWIR) and long-wave infrared (LWIR) detection offers a compelling alternative to conventional infrared detecto...

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Inequalities in the distribution of the family physician workforce in the kingdom of Saudi Arabia: a regional analysis

Mohammed Khaled Al-Hanawi 2026-11-01 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Although the national distribution of family physicians appears balanced, Lorenz curve analysis reveals significant inequalities across regions, sectors, and demographic groups. These findings have implications for workforce planning and monitoring, particularly in regions exhibiting co...

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Assessing effects of low-dose CT attenuation correction on quantification and image quality on a LAFOV PET system with a modified NEMA-IQ phantom

👤 Paul K Marsden 📅 2026-11-02 📖 1 min 📝 77 words LOW VISION

Summary: CONCLUSIONS: Even for low-dose CT protocols, PET quantification remained consistent in all NEMA-IQ spheres for many materials. Beam hardening and photon starvation in the CT image are likely to occur in specific body regions such as hips, shoulders or the abdomen for an arms-down scan and these effe...

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Functional and gain-based vestibulo-ocular measures across serum 25(OH)D categories: a dissociation between fHIT performance and vHIT gain

👤 Alper Köycü 📅 2026-11-02 📖 1 min 📝 30 words LOW VISION

Summary: CONCLUSIONS AND SIGNIFICANCE: Low 25(OH)D was accompanied by reduced functional vestibulo-ocular performance in the vertical canal planes despite normal gains. fHIT may capture subclinical change that vHIT gain does not.

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HiCP2GAN: a plug and play foundation model-based GAN for Hi-C enhancement

👤 Oluwatosin Oluwadare 📅 2026-11-02 📖 1 min 📝 65 words LOW VISION

Summary: The 3D organization of chromatin shapes gene regulation and cellular function, and Hi-C has become the primary technique for mapping chromatin interactions genome-wide. However, high-resolution data remain costly and scarce, leaving many studies with sparse contact maps that limit downstream analysi...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

👤 Alan J Emanuel
📅 2026-10-25 📖 1 min 📝 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

Felix Blankenburg 2026-10-30 1 min 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

Hamdan Hamdan 2026-10-30 1 min 33 words TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Dosage and safety of non-invasive brain stimulation for chronic musculoskeletal pain management: a scoping review

👤 Liming Jiang
📅 2026-10-30 📖 1 min 📝 64 words

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Summary: CONCLUSION: Current NIBS dosage for management of CMP remains heterogeneous and driven by empirical experience. Safety reporting is inadequate. Although reported adverse events were predominantly mild and transient, inconsistent adverse event ascertainment and reporting preclude reliable conclusion ...

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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https://pubmed.ncbi.nlm.nih.gov/42962808/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102172624&v=2.20.1

EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42962828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102172624&v=2.20.1

Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42963327/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102172624&v=2.20.1

Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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https://pubmed.ncbi.nlm.nih.gov/42964054/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102172624&v=2.20.1

Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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https://pubmed.ncbi.nlm.nih.gov/42964133/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102172624&v=2.20.1

Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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https://pubmed.ncbi.nlm.nih.gov/42964233/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102172624&v=2.20.1

Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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https://pubmed.ncbi.nlm.nih.gov/42964313/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102172624&v=2.20.1

Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

🔗 **Read full article:**
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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42966015/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102172624&v=2.20.1

State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu | 2026-11-01 | 1 min | 46 words | BRAIN COMPUTER INTERFACE

> **Summary:** CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42966748/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102172624&v=2.20.1

CPython core dev Larry Hastings on testing race conditions in free-threaded Python and AI-coding

/u/miabajic | 2026-11-02 | 1 min | 239 words | REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <p>Hey everyone! In my free time I host Behind the Commit, a podcast with open-source maintainers. I wanted to talk to Larry Hastings because free-threaded Python brought lots of challenges with testing race conditions -- I myself had no idea how to test code without G...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xm18lx/cpython_core_dev_larry_hastings_on_testing_race/

FastAPI Cloud is now generally available

👤 /u/tiangolo 📅 2026-11-02 📖 1 min 📝 68 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I'm excited to share with you all that FastAPI Cloud is now generally available: https://fastapicloud.com/blog/fastapi-cloud-generally-available/ <hr /> Note: I thought it could be...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xm7oif/fastapi_cloud_is_now_generally_available/

Skill to Create Product Videos

👤 aaitor 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/aaitor/skill-video-studio Comments URL: https://news.ycombinator.com/item?id=50340992 Points: 44 # Comments: 14

🔗 **Read full article:**
https://github.com/aaitor/skill-video-studio

Sneaking into the auditorium for the Windows 2000 launch rehearsal

👤 ibobev 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://devblogs.microsoft.com/oldnewthing/20261102-00/?p=112818/ Comments URL: https://news.ycombinator.com/item?id=50341028 Poin...

🔗 **Read full article:**
https://devblogs.microsoft.com/oldnewthing/20261102-00/?p=112818/

Astra can read and write weights to improve Qwen

👤 dbreunig 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://rikiyat.github.io/blog/weight-space-programming/ Comments URL: https://news.ycombinator.com/item?id=50341174 Points: 1395 # ...

🔗 **Read full article:**
https://rikiyat.github.io/blog/weight-space-programming/

Agentic backdoors via pretraining poisoning

👤 gmays 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://arxiv.org/abs/2610.37741 Comments URL: https://news.ycombinator.com/item?id=50341334 Points: 71 # Comments: 19

🔗 **Read full article:**
https://arxiv.org/abs/2610.37741

Ubuntu still has not released patch for Linux 0day RCE in almost 3 weeks

👤 majorchord 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://freenode.net/article/ubuntu-lts-kernels-still-unfixed-for-critical-cve-2026-98432 Comments URL: <a href="https://news.ycombinator.com/item?id=50341462">https://news....

🔗 **Read full article:**
https://freenode.net/article/ubuntu-lts-kernels-still-unfixed-for-critical-cve-2026-98432

Ballot Rejections Have Soared in Rural Counties as USPS Reduces Mail Service

👤 ck2 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.propublica.org/article/mail-ballot-delays-postal-service-midterms Comments URL: <a href="https://news.ycombinator.com/item?id=50341616">https://news.ycombinator.com/item?id=...

🔗 **Read full article:**
https://www.propublica.org/article/mail-ballot-delays-postal-service-midterms

Egoless Mathematics in the AI Era

👤 smilelamp 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://terrytao.wordpress.com/2026/11/02/egoless-mathematics-in-the-ai-era/ Comments URL: <a href="https://news.ycombinator.com/item?id=50341736">https://news.ycombinator.com/item?id=50...

🔗 **Read full article:**
https://terrytao.wordpress.com/2026/11/02/egoless-mathematics-in-the-ai-era/

Oracle files to sell $20B in new stock, bonds to finance AI buildout

👤 smartmic 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.cnbc.com/2026/11/02/oracle-files-to-sell-20-billion-in-new-stock-bonds-to-finance-ai-buildout.html Comments URL: <a href="https://news.ycomb...

🔗 **Read full article:**
https://www.cnbc.com/2026/11/02/oracle-files-to-sell-20-billion-in-new-stock-bonds-to-finance-ai-buildout.html

BYD sold more EVs last month than the US did all quarter, topping 300,000

👤 cisc 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://electrek.co/2026/11/02/byd-sold-more-evs-last-month-than-the-us-did-all-quarter/ Comments URL: <a href="https://news.ycombinator.com/item?id=50344181">https://news.yc...

🔗 **Read full article:**
https://electrek.co/2026/11/02/byd-sold-more-evs-last-month-than-the-us-did-all-quarter/

Can't fish Ethernet? Use "invisible" fiber instead

👤 _mpn_ 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://momandpop.network/2026/11/02/cant-fish-ethernet-use-invisible-fiber-instead/ Comments URL: <a href="https://news.ycombinator.com/item?id=50344222">https://news.ycombinato...

🔗 **Read full article:**
https://momandpop.network/2026/11/02/cant-fish-ethernet-use-invisible-fiber-instead/

There are 13 ways to analyse a replication study, but only one of them is coherent.

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-05-03 📖 6 min 📝 1361 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25 30 min 6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3434 words

TWENTY PERCENT STATISTICIAN

Summary: Update 28-10-2026: ResearchBox now has an API. This new information has been incorporated in the text below. The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Sci...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

👤 Adriel Carridice 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/episode-29-qa-with-dr-anahita-kyani-director-of-neuroscience-data-science-and-digital-health-at-johnson-johnson-innovative-medicine/

Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-felix-schwock-phd-student-at-university-of-washington-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Neuroethical Perspectives on Neurotechnology in Latin America

Adriel Carridice 2026-09-23 1 min 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

Deidre Artis 2026-07-17 1 min 34 words EMBS

Summary: Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

🔗 **Read full article:**
https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/

A prospective randomized study of patient-centered AI for clinical visit preparation

👤 Xincheng Tan, Matthew Thompson, Xiang Ji, Angelica Willis, Will Vaughan, Roma Ruparel, Michael A. Sanchez, Sean Chang, Joseph Xu, Bhavna Daryani, Preeti Singh, Pinal Bavishi, Kejia Chen, Fan Zhang, Priya Gupta, Ernest Lim, Christina Chen, Bob Lou, Dimitrios Antos, Zach Wasson, Ray Luo, Fadi Yousif, Florence Thng, Mark Malhotra, Joseph Breda, Daniel McDuff, Kavi Goel, Katherine Chou, Dale R. Webster, Avinatan Hassidim, Yossi Matias, Po-Hsuan Cameron Chen, Yun Liu, Mike Schaekermann, Quang Duong, Jackie M. Barr

📅 2026-11-02 📖 1 min 📝 282 words ARXIV CS HC

Summary: arXiv:2610.41437v1 Announce Type: new Abstract: Patients often arrive at brief, time-constrained clinical visits unsure of what to expect or ask, and millions of adults now turn to general-purpose AI chatbots to prepare beforehand. Yet whether conversational AI improves visit readiness remains uncl...

🔗 **Read full article:**
https://arxiv.org/abs/2610.41437

Hybrid Computer Use: A Unified Programmatic, Semantic, and Visual Interface for General Agents

Haoqing Wang, Xiaoyi Zhang, Sicheng Fan, Zhexin Zhang, Xinjie Zhang, Jiahao Li, Bin Li, Yan Lu

2026-11-02 1 min 212 words ARXIV CS HC

Summary: arXiv:2610.41432v1 Announce Type: new Abstract: Computer-use agents increasingly tackle tasks ranging from everyday desktop activities to long-horizon professional workflows. Yet interfaces centered on screenshots and mouse-and-keyboard actions can impose a human-interaction prior, requiring repeat...

Read full article:
https://arxiv.org/abs/2610.41432

When History Overrides Intent: Relational Schema Lock-In in Human-LLM Interaction

Wei Yang

2026-11-02 1 min 148 words ARXIV CS HC

Summary: arXiv:2610.41367v1 Announce Type: new Abstract: Long-context interaction and persistent memory allow LLM assistants to develop history-grounded understandings of users. When users change direction, however, an understanding built through earlier exchanges may conflict with their current instruction...

Read full article:
https://arxiv.org/abs/2610.41367

Toward Low-Burden Image-Based Dietary Assessment Through Human-AI Collaborative Correction

Darren Liu, Andrew Lu, Zhiyuan "Ryan" Wang, Hanqi Luo, Xiao Hu, Runze Yan

2026-11-02

1 min

152 words

ARXIV CS HC

Summary: arXiv:2610.41300v1 Announce Type: new Abstract: Image-based dietary assessment can automate initial food logging, but meal photographs often leave food identity, hidden ingredients, preparation, and portion uncertain, making correction a recurring part of use. We introduce Human-AI Collaborative Co...

Read full article:
https://arxiv.org/abs/2610.41300

Imputing Missing Data in Longitudinal Life-Logging Athlete Data

Saaz Kaur Sahdra, Kalina Yacef, HaeJin Lee, Judy Kay

2026-11-02

1 min

187 words

ARXIV CS HC

Summary: arXiv:2610.41226v1 Announce Type: new Abstract: Long-term, diverse personal informatics data enables athletes to understand and act on their health and performance goals, whether assessing readiness right now, tracking how soreness relates to training load over time, or predicting how sleep affects...

Read full article:
https://arxiv.org/abs/2610.41226

Flash Word : ESL Listening Support with Subtitle Skimming

Yuzu Saijo, Jun Rekimoto
2026-11-02 1 min 131 words ARXIV CS HC

Summary: arXiv:2610.41098v1 Announce Type: new Abstract: Listening to English audio with English subtitles is thought to be effective for ESL students to improve listening comprehension skills. However, for those who learn English as a second language it can be hard to finish English subtitles. Also they ar...

🔗 **Read full article:**
https://arxiv.org/abs/2610.41098

Out of sync: Accommodation, evaluation, and emergence in five weeks with an agentic AI teammate

Nia Nixon, Jaeyoon Choi, Pedro Martins de Bastos, Mohammad Amin Samadi
2026-11-02

1 min 216 words ARXIV CS HC

Summary: arXiv:2610.41057v1 Announce Type: new Abstract: As agentic AI in education shifts from reactive tools toward partners that take initiative and persist over time, a central question is how learners relate to, evaluate, and benefit from artificial agency. We conducted a longitudinal experiment in an ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.41057

Measuring Personal Information Exposure in Mental Health-Related Conversations with LLMs

Synthia Qia Wang, Ranya Sharma, Christine Lee, Sarah Keedy, Marshini Chetty, Nick Feamster

2026-11-02 1 min 155 words ARXIV CS HC

Summary: arXiv:2610.41045v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly used for informal mental health support, yet the privacy implications of conversational interaction remain poorly understood. Prior LLM privacy research has focused largely on model-side risks such as memo...

Read full article:
https://arxiv.org/abs/2610.41045

Robot Navigation and Canine Welfare: A Pilot Study Surfacing Challenges for Cross-Species Legibility and Proxy Validity in Animal-Machine Interaction

Haoxiang Zhang, Nancy Smith

2026-11-02 1 min 159 words ARXIV CS HC

Summary: arXiv:2610.40921v1 Announce Type: new Abstract: Autonomous delivery robots increasingly share residential spaces with companion animals, yet navigation behavior is rarely designed with animal welfare in mind. This paper presents a pilot study evaluating a curve-out maneuver, a laterally arcing pass...

Read full article:
https://arxiv.org/abs/2610.40921

Personalised 3D Mnemonics for Vocabulary Learning in Context

👤 Karolina Trajkovska, Matja\v{z} Kljun, Maheshya Weerasinghe, Klen \v{C}opi\v{c} Pucihar

📅 2026-11-02 📖 1 min 📝 185 words ARXIV CS HC

Summary: arXiv:2610.40760v1 Announce Type: new Abstract: Augmented reality (AR) can support vocabulary learning by placing mnemonic cues in the physical contexts where new words are encountered. Prior work has augmented real objects with bilingual labels, audio, keywords, and 3D visualisations of those keyw...

🔗 **Read full article:**
https://arxiv.org/abs/2610.40760

Critical Flicker Fusion Frequency As An Experience-Restricted Constraint On Visual Temporal Resolution: What Does And Does Not Change It

Natalia D. Rydzenska, Pawel J. Winklewski, Michal W. Blaszczyk-Niezgoda, Anna B. Marcinkowska

2026-11-02 1 min 250 words ARXIV QBIO NC

Summary: arXiv:2607.29068v3 Announce Type: replace Abstract: Experience-dependent plasticity is fundamental to adaptive behaviour, yet the conditions under which basic sensory timing can be modified in adulthood remain poorly specified. Critical flicker fusion frequency (CFFF), the threshold at which flicke...

Read full article:
https://arxiv.org/abs/2607.29068

Perceived vertical and eye level as one orientation process: a closed-form account of the Li-Matin combination rules for egocentric space

A. Y. Shavit

2026-11-02 1 min 302 words ARXIV QBIO NC

Summary: arXiv:2607.19681v5 Announce Type: replace Abstract: Li and Matin (2005a, 2005b) reported three regularities of the induced visually-perceived vertical (VPV), the direction judged upright. For small tilts, a peripheral line shifts it nearly linearly with tilt, two lines combine nearly linearly, and ...

Read full article:
https://arxiv.org/abs/2607.19681

Message interaction dynamics covary with brain volume across mammalian connectomes

Yan Hao, Hannah Lax, Marc-Thorsten H\"utt, Daniel J. Graham

2026-11-02 1 min

197 words

ARXIV QBIO NC

Summary: arXiv:2505.15477v3 Announce Type: replace Abstract: Brain network communication models typically assume that signals propagate independently, despite the high network density and small diameter of mammalian connectomes, where interactions among simultaneously propagating messages are likely. We inv...

Read full article:
https://arxiv.org/abs/2505.15477

Building functional and mechanistic models of cortical computation based on canonical cell type connectivity

Arno Granier, Katharina A Wilmes, Mihai A Petrovici, Walter Senn

2026-11-02 1 min

154 words

ARXIV QBIO NC

Summary: arXiv:2504.03031v3 Announce Type: replace Abstract: Neuronal circuits of the cerebral cortex are the structural basis of mammalian cognition. The same qualitative components and connectivity motifs are repeated across functionally specialized cortical areas and mammalian species, suggesting a singl...

Read full article:
https://arxiv.org/abs/2504.03031

Lattice-Based Pruning of Recurrent Neural Networks via Poset Modeling: A Corrected Theory and a Matched-Sparsity Evaluation

Rakesh Sengupta

2026-11-02 1 min 197 words ARXIV QBIO NC

Summary: arXiv:2502.16525v2 Announce Type: replace Abstract: Magnitude pruning of recurrent neural networks (RNNs) ignores how information flows between neurons. An earlier version proposed to model RNN dependencies as a partially ordered set (poset) and to retain the meet-irreducible elements of its depend...

🔗 **Read full article:**
https://arxiv.org/abs/2502.16525

Explainable AI-Driven Neural Activity Analysis in Parkinsonian Rats under Electrical Stimulation

Jibum Kim, Hanseul Choi, Gaeun Kim, Sunggu Yang, Eunha Baeg, Donggue Kim, Seongwon Jin, Sangwon Byun

2026-11-02 1 min 167 words ARXIV QBIO NC

Summary: arXiv:2502.12471v2 Announce Type: replace Abstract: Parkinson's disease (PD) is characterized by motor dysfunction and abnormal neural oscillations. This study combined graphene-based cortical stimulation with an explainable artificial intelligence (XAI) framework to investigate spatial and frequen...

🔗 **Read full article:**
https://arxiv.org/abs/2502.12471

Hyperbolic embedding of higher-order simplicial networks

Martin Guillemaud, Vito Latora, Mario Chavez

2026-11-02 1 min 177 words

ARXIV QBIO NC

Summary: arXiv:2610.41684v1 Announce Type: new Abstract: Many biological interactions involve groups rather than pairs, as when several proteins assemble into a multi-protein complex. Simplicial complexes naturally capture these collective interactions. Existing hyperbolic embeddings represent network hiera...

🔗 **Read full article:**
https://arxiv.org/abs/2610.41684

Expression of White and Gray Matter Autism Functional Network Connectivity Signatures in Infancy

Qiang Li, Jingyu Liu, Zening Fu, Armin Iraji, Sarah Shultz, Vince D. Calhoun

2026-11-02 1 min 211 words

ARXIV QBIO NC

Summary: arXiv:2610.40839v1 Announce Type: new Abstract: Functional connectivity studies have provided valuable insights into altered brain organization in autism spectrum disorder (ASD). However, most functional connectivity studies focus exclusively on either gray matter (GM) regions or white matter (WM) ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.40839

Prolonged visual-motor dual-task effects on dynamic stability in children with ADHD subtypes

📅 2026-11-02 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-77122-3

DIA proteomics identifies time-of-day-dependent hippocampal proteomic signatures associated with exercise timing in type 2 diabetic mice

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Factors influencing plasma brain-derived neurotrophic factor (BDNF) in adolescent sport participants

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Association of diffusion along the perivascular spaces with clinical symptoms and dopaminergic degeneration in Parkinson’s disease

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Genomics for chronic overlapping pain conditions

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A multi-institutional investigation of psilocybin’s effects on mouse behavior

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State space models naturally produce time cell and oscillatory behaviors and scale to abstract cognitive functions

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Tissue-specific lipidomic alterations in aging rats

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Microglia-dependent PDGFB regulation of neuronal KCND2 in the hypothalamic paraventricular nucleus determines propofol anesthesia induction

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Author Correction: mTORC1 drives cell-autonomous astrocyte reactivity in tuberous sclerosis

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https://www.nature.com/articles/s41586-026-11350-5

Serotonin signaling enhances phagocytosis by brain macrophages through Htr2b and alleviates brain injury after stroke

👤 Peiying Li 📅 2026-10-28 📖 1 min 📝 45 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 28 October 2026; doi:10.1038/s41593-026-02482-y The serotonin receptor Htr2b activation enhances lysosome biogenesis and phagocytic clearance of debris to mitigate neuroinflammation and brai...

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https://www.nature.com/articles/s41593-026-02482-y

A multi-institutional investigation of psilocybin’s effects on mouse behavior

Robert C. Malenka
2026-11-02 1 min 44 words NATURE NEUROSCIENCE

> **Summary:** Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02500-z In a study across five independent labs, 2 mg kg−1 psilocybin in mice acutely increased avoidance-related behaviors in a robust, reproducible...

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https://www.nature.com/articles/s41593-026-02500-z

Bite-block effects on native s/ʃ and English r/l in child speakers of a language with/without this contrast

F. Sayako Earle
2026-11-02 1 min 171 words FRONTIERS HUMAN NEUROSCIENCE

> **Summary:** Learning to produce non-native speech contrasts requires developing new phonological categories as well as new articulatory-perceptual mappings for these new categories. However, within the same speakers, it remains unclear how these new categories compare to contrasts that are already fully establi...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1949253

From biomechanical data to robotic execution: design and control of an intelligent Tuina system

Wei Yao 2026-11-02 1 min 181 words FRONTIERS NEUROROBOTICS

Summary: IntroductionTraditional Chinese medicine (TCM) Tuina (massage) relies heavily on practitioners' physical stamina and subjective experience, leading to operational variability that hinders therapeutic standardization and scalable rehabilitation service delivery. To address this limitation, this study...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1937295

Mixed-timestep spiking neural networks: a temporal alignment framework for high-efficiency neuromorphic computing

Liang Chen 2026-11-02 1 min 229 words FRONTIERS NEUROSCIENCE

Summary: Spiking Neural Networks (SNNs) are energy-efficient due to their event-driven computation and sparse spike-based communication. However, most existing SNN models adopt a uniform time-step across all layers, which limits temporal flexibility and leads to suboptimal accuracy latency trade-offs under l...

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Evidence for decoder-label-regime-dependent effects of complementary leaky integrate-and-fire temporal-state decoding in motor-imagery EEG

Dong Zhang 2026-11-02 1 min 272 words FRONTIERS NEUROSCIENCE

Summary: BackgroundMotor-imagery electroencephalography often provides limited subject-specific supervision. This study tested whether complementary leaky integrate-and-fire (CLIF)-family temporal-state decoding shows a decoder-level advantage after the EEG representation is fixed, how that advantage changes...

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Diurnal cortisol, perceived sleep quality, actigraphic sleep and depression: an exploratory pilot in patients with head and neck cancer

Sandra E. Sephton 2026-11-02 1 min 231 words FRONTIERS NEUROSCIENCE

Summary: BackgroundIn patients with head and neck cancer (HNC), perceived sleep quality, actigraphy-derived sleep parameters, depressive symptoms, and diurnal cortisol may represent related behavioral and biological processes. This exploratory study tested a priori associations among these constructs.Methods...

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Case Report: Clinical improvement of localized chronic eczema following multisegmental paraspinal electroacupuncture

Jianhua Tang 2026-11-02 1 min 172 words FRONTIERS NEUROSCIENCE

Summary: BackgroundNeuroimmune interactions provide a rationale for studying somatosensory neuromodulation in inflammatory disorders, but a single clinical observation cannot establish an autonomic mechanism. We report a case of localized chronic eczema that improved after multisegmental paraspinal electroac...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1905399

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Intramedullary Spinal Tuberculosis Mimicking Neoplasm with Ocular Involvement in an Adolescent: A Case Report

👤 Indri Febriani 📅 2026-10-31 📖 1 min 📝 72 words LOW VISION

Summary: Tuberculosis (TB) continues to represent one of the key health problems of high-burden nations. The childhood population is particularly vulnerable to extrapulmonary TB. Spinal intramedullary TB is extremely uncommon and can imitate neoplastic conditions, leading to a late diagnosis with fatal outco...

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Framework-informed analysis of practice: A qualitative study on climate resilience in primary care in Austria

Katharina Brugger 2026-10-31 1 min 59 words LOW VISION

Summary: CONCLUSIONS: PCPs play a vital role in local climate resilience but require tailored, context-specific guidance aligned with PC's integrated, small-scale, multiprofessional nature. Strategic macro-level objectives are required to support low-barrier implementation of resilience measures. Prioritizin...

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Early applications of apple vision pro in clinical medicine and medical education: a scoping review

Chuxu Wang 2026-10-31 1 min 58 words LOW VISION

Summary: The Apple Vision Pro (AVP), a next-generation spatial computing device, has demonstrated potential applications in multidisciplinary clinical diagnosis, treatment, and medical education, by leveraging its unique technical architecture and performance advantages. In clinical practice, its fundamental...

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A Distributed Vision Transformer-Based Deep Learning Framework for Early Detection and Classification of Lumbar Disc Degeneration

👤 N Chaitanya Kumar
📅 2026-10-31 📖 1 min 📝 68 words

LOW VISION

Summary: Lumbar disc degeneration (LDD) is a principal structural cause of chronic low back pain. Magnetic resonance imaging (MRI) is the reference modality for assessing the intervertebral discs, but severity grading is performed manually, is limited by inter-observer variability, and becomes a bottleneck a...

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A Result-Auditable Protocol for Low-Data Classification of Blue-and-White Porcelain Botanical Motifs Using Frozen Chinese-CLIP Features

👤 Chenghao Yu
📅 2026-10-31 📖 1 min 📝 58 words

LOW VISION

Summary: Low-data visual classification is common in cultural heritage documentation, where expert labels are expensive, and image sources are heterogeneous. This protocol provides result-level audit support for classifying five botanical motifs-peony, lotus, chrysanthemum, plum blossom, and bamboo-on blue-a...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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All-optical 2D perovskite photodetector arrays for pixelated long-wave infrared imaging

Peijun Guo 2026-11-01 1 min 55 words LOW VISION

Summary: Infrared detection and imaging are important for applications including thermal monitoring, night vision, environmental sensing, and molecular identification. All-optical mid-wave infrared (MWIR) and long-wave infrared (LWIR) detection offers a compelling alternative to conventional infrared detecto...

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Inequalities in the distribution of the family physician workforce in the kingdom of Saudi Arabia: a regional analysis

Mohammed Khaled Al-Hanawi 2026-11-01 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Although the national distribution of family physicians appears balanced, Lorenz curve analysis reveals significant inequalities across regions, sectors, and demographic groups. These findings have implications for workforce planning and monitoring, particularly in regions exhibiting co...

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Assessing effects of low-dose CT attenuation correction on quantification and image quality on a LAFOV PET system with a modified NEMA-IQ phantom

Paul K Marsden 2026-11-02 1 min 77 words LOW VISION

Summary: CONCLUSIONS: Even for low-dose CT protocols, PET quantification remained consistent in all NEMA-IQ spheres for many materials. Beam hardening and photon starvation in the CT image are likely to occur in specific body regions such as hips, shoulders or the abdomen for an arms-down scan and these effe...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel
2026-10-25 1 min 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

Felix Blankenburg 2026-10-30 1 min 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

Hamdan Hamdan 2026-10-30 1 min 33 words TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Dosage and safety of non-invasive brain stimulation for chronic musculoskeletal pain management: a scoping review

👤 Liming Jiang
📅 2026-10-30 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: CONCLUSION: Current NIBS dosage for management of CMP remains heterogeneous and driven by empirical experience. Safety reporting is inadequate. Although reported adverse events were predominantly mild and transient, inconsistent adverse event ascertainment and reporting preclude reliable conclusion ...

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

Chuhuai Wang
2026-10-31 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

👤 Taomei Guo 📅 2026-10-31 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

👤 Isao Matsushita 📅 2026-10-31 📖 1 min 📝 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

Shouyan Wang 2026-10-30 1 min 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

Aishi Zhou 2026-10-30 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Implicit learning of melodic structure: A role for pitch?

2024-01-22 1 min 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000303

To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 687 words FMHY

> **Summary:** I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

From Charlemagne to the Kitchen Boy: Reading the Hamburg Rathaus

👤 mertbio 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://alsterrunde.com/from-charlemagne-to-the-kitchen-boy-reading-the-hamburg-rathaus/ Comments URL: <a href="https://news.ycombinator.com/item?id=50333774">https://news.yc...

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Show HN: HackerHack – HackerBooks but for Posts Posted on HN

👤 mw888 📅 2026-11-02 📖 1 min 📝 17 words HACKER NEWS

> **Summary:** No AI used for this btw. <hr /> Comments URL: https://news.ycombinator.com/item?id=50333877 Points: 77 # Comments: 14

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China's First Frontier AI Export Controls: Mofcom Announcement No. 48

👤 jonnybgood 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://labs.cloudsecurityalliance.org/research/csa-research-note-china-ai-export-controls-announcement-48-2/ Comments URL: <a href="https://news.ycombin...

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BioWare co-founder: game execs "have never made a game" and don't get it

👤 entryleveljobsc 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** <p>Article URL: <a href="https://www.pcgamer.com/gaming-industry/none-of-these-people-have-ever-made-a-game-bioware-co-founder-greg-zeschuk-says-of-game-industry-executives-they-just-dont-know-how-it-works/">https://www.pcgamer.com/gaming-industry/none-of-these-people-have-ever-made-a-game-bioware-c...

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Show HN: I've updated my Linux System Call website and API

👤 guitmz 📅 2026-11-02 📖 1 min 📝 57 words HACKER NEWS

> **Summary:** <p>I first released those in back in 2020 and now I finally added more architectures, Kernel selector, syscall descriptions and a few other niceties. It should work a bit better on mobile devices too but I am no CSS expert.<p>The API is at <a href="https://api.syscall.sh" rel="nofollow">https://api....

🔗 **Read full article:**
https://syscall.sh/

The future in old computer magazines (1984–2004)

👤 tudell 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://tomorrowasusual.substack.com/p/blast-from-the-past-the-future-in Comments URL: <a href="https://news.ycombinator.com/item?id=50335341">https://news.ycombinator.com/item?id=50335341</...

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https://tomorrowasusual.substack.com/p/blast-from-the-past-the-future-in

Supermarket freezers in Belgium and Sweden hit by suspected cyber-attack

👤 xqcgrek2 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.bbc.com/news/articles/cq1kd1nee80eo Comments URL: https://news.ycombinator.com/item?id=50335602 Points: 52 # Comments: 48

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https://www.bbc.com/news/articles/cq1kd1nee80eo

Supermarket freezers across Europe hit by second suspected cyber-attack

👤 fergie 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.theguardian.com/world/2026/nov/02/supermarket-freezers-across-europe-hit-by-second-suspected-cyber-attack Comments URL: <a href="http...

🔗 **Read full article:**
https://www.theguardian.com/world/2026/nov/02/supermarket-freezers-across-europe-hit-by-second-suspected-cyber-attack

Never Complain, Never Explain

👤 gregsadetsky 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://en.wikipedia.org/wiki/Never_complain,_never_explain Comments URL: https://news.ycombinator.com/item?id=50335659 Points: 4 <p...

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https://en.wikipedia.org/wiki/Never_complain,_never_explain

Private Jet plane crashes in Florida

👤 hmokiguess 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.cbc.ca/news/world/florida-plane-crash-9.7404996 Comments URL: https://news.ycombinator.com/item?id=50335914 Points: 6 <p...

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https://www.cbc.ca/news/world/florida-plane-crash-9.7404996

T-MRB Call for Nominations for Associate Editors

👤 Deidre Artis 📅 2026-04-22 📖 1 min 📝 16 words EMBS

> **Summary:** The post T-MRB Call for Nominations for Associate Editors appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/t-mrb-call-for-nominations-ae/#new_tab

2026 Adcom Petition is Open

👤 Deidre Artis 📅 2026-09-11 📖 1 min 📝 14 words EMBS

> **Summary:** The post 2026 Adcom Petition is Open appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/about/embs-officers/2026-embs-administrative-committee-adcom-election-2/#new_tab

Analysis of the dynamic evolution and immunosuppressive functions of myeloid cells in gliomas based on single-cell transcriptomics

📖 1 min 📝 23 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Hao Yuan, Jingyi Yang, Zili Bai, Hongyan Wang, Jiaxuan Tong, Peng Chen, Shouping Gong

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S000689932600466X?dgcid=rss_sd_all

Common phenomenal and neural substrate geometry in visual motion perception

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Kallum Robinson, Ariel Zeleznikow-Johnston, Jiahao Wu, Yumiko Yoshimura, Naotsugu Tsuchiya

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006282?dgcid=rss_sd_all

Neuronal and glial transcriptional states vary with proximity to tangle-bearing neurons in the human entorhinal cortex

👤 Boulger, S. L., Lambie, N., Wos, J., Papageorgopoulou, M., Mendez Pastor, S., Adair, E., Thomas, M., Caramello, A., Fancy, N. N., Jackson, J. S., Duff, E. P., Mallach, A., Matthews, P. M.

📅 2026-11-01 📖 1 min 📝 150 words BIORXIV NEUROSCIENCE

Summary: More neurons die in Alzheimer's disease than form tangles, and tangle-bearing neurons can persist for decades, so tangles alone cannot explain degeneration. We combined 6,000-plex spatial transcriptomics with PHF1 immunofluorescence (167,176 cells; 9 donors) and imaging mass cytometry (170,195 cells...

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https://www.biorxiv.org/content/10.64898/2026.10.27.762234v1?rss=1

Microglial NUS1 deficiency drives dopaminergic neurodegeneration through intercellular iron dysregulation in Parkinson’s disease models

Haigang Ren 2026-10-26 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 26 October 2026; doi:10.1038/s41593-026-02499-3 Using cellular and mouse models, this study reports that deficiency of the Parkinson’s disease risk gene NUS1 in microglia disrupts intercellu...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02499-3

Chronic progressive demyelination induces a microglial state with lysosomal dysfunction

Mikael Simons 2026-10-26 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 26 October 2026; doi:10.1038/s41593-026-02489-5 Acute myelin injury responses are well defined, but microglia adaptations to chronic damage remain unclear. Mouse models reveal a chronic demy...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02489-5

Calcium permeability–enhancing mutations in GluRIIC suppress sustained presynaptic homeostatic plasticity triggered by loss of GluRIIA

Olfat A. MalakLindsay GrayAmmar AlyRobin W. BallJessica Gustin-MurrayMyriam MoujahidineGuadalupe CarrascoAndrew RoskoEhud Y. IsacoffA. Pejmun Haghighiahttps://ror.org/050sv4x28Buck Institute for Research on Aging, Novato, CA 94945bhttps://ror.org/04b6nzv94Brigham and Women's Hospital, Harvard Medical School, Boston, MA 02115cInscopix Inc., Mountain View, CA 94043dhttps://ror.org/03taz7m60Leonard Davis School of Gerontology, University of Southern California, Los Angeles, CA 90089ehttps://ror.org/01pxwe438Department of Physiology, McGill University, Montreal, QC H3G1Y6, Canadafhttps://ror.org/01an7q238Helen Wills Neuroscience Institute, University of California, Berkeley, CA 94720ghttps://ror.org/01an7q238Department of Neuroscience, University of California, Berkeley, CA 94720hhttps://ror.org/01an7q238Department of Molecular and Cell Biology, University of California, Berkeley, CA 94720

📅 2026-10-27 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAccumulating experimental evidence indicates that neurons adjust their synaptic strength, in response to genetic or environmental/ pharmacological perturbations, to ensure stability in neural circu...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625954123?af=R

VAP organizes presynaptic Ca2+ signaling and synaptic vesicle mobilization through Kv2 membrane contact sites

👤 Cameron D. PatonIn Ha ChoMichael B. Hoppaahttps://ror.org/049s0rh22Division of Arts and Sciences, Department of Biological Sciences, Dartmouth College, Hanover, NH 03755

📅 2026-10-27 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThe endoplasmic reticulum (ER) supports neuronal function by regulating Ca2+, lipid, and protein homeostasis, yet how ER–plasma membrane contact sites contribute to synaptic transmission remains p...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2602106123?af=R

Intraventricular clindamycin enhances interstitial brain fluid drainage and the blood–brain barrier after a traumatic brain injury

👤 Landon S. AshbyMegan G. CroweDaveraj ChakravartySanjeev MedamGabriel FramptonAyesha B. DasDavid A. HormuthThomas E. YankeelovVincent Y. C. WangLinda J. Noble-HaeussleinBenjamin J. Umlaufahttps://ror.org/00hj54h04Department of Neurosurgery, The University of Texas at Austin, Austin, TX 78712bhttps://ror.org/00hj54h04Department of Biomedical Engineering, The University of Texas at Austin, Austin, TX 78712chttps://ror.org/00hj54h04Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin, Austin, TX 78712dhttps://ror.org/00hj54h04Cancer Research Center, The University of Texas at Austin, Austin, TX 78712ehttps://ror.org/00hj54h04Department of Diagnostic Medicine, The University of Texas at Austin, Austin, TX 78712fDepartment of Imaging Physics, The University of Texas M.D. Anderson Cancer Center, Houston, TX 77030ghttps://ror.org/00hj54h04Department of Neurology, The University of Texas at Austin, Austin, TX 78712hhttps://ror.org/00hj54h04Department of Psychology, The University of Texas at Austin, Austin, TX 78712ihttps://ror.org/00hj54h04Mulva Clinic for the Neurosciences, The University of Texas at Austin, Austin, TX 78712

📅 2026-10-27 📖 1 min 📝 44 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDespite advances in the acute clinical management of severe traumatic brain injuries, treating injury-derived secondary pathogenesis remains a major challenge. We describe an alternative strategy ...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2536755123?af=R

Digital health meets Parkinson’s disease: a bibliometric exploration of wearable sensors research and emerging frontiers (2005–2026)

Lan Xiong
2026-10-08 1 min 228 words FRONTIERS NEUROINFORMATICS

Summary: BackgroundWearable and sensor-based monitoring technologies are increasingly used to support objective, longitudinal assessment in Parkinson’s disease (PD). However, the global development, knowledge structure, and emerging research frontiers of this interdisciplinary field remain incompletely mappe...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1937585

MTDANet: lightweight multi-scale temporal hierarchy and distance-regularized linear attention network for motor imagery brain–computer interfaces

Binbin Chen, Chen Deng, Huanmin Ge, Xinhua Su and Yanfei Shen
2026-10-26 1 min 239 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Motor-imagery (MI) electroencephalography (EEG) offers a non-invasive route to brain–computer interaction, yet reliable single-trial decoding remains difficult because task-related patterns are weak, nonstationary, and distributed over multiple temporal scales. This study investigates whe...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb330

AOR: adaptive orthogonal regression for depression feature selection with incomplete EEG channels

👤 Wenjia Dong, Zhijian Gong, Xueyuan Xu, Ke Liu, Fulin Wei and Li Zhuo

📅 2026-10-26 📖 1 min

📝 272 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Depression induces severe mental and physical impairments, highlighting the clinical need for rapid and reliable detection. Electroencephalography (EEG) directly reflects neural electrical activity and serves as a promising tool for depression investigation. However, the inherent non-stat...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb32f

Multimodal regulation of neurite extension by extracellular matrix composition and electrical stimulation on aligned collagen fiber scaffolds

👤 Madalynn J Thompson, Joanna Borowska-Fielding, Ivy A Murphy, Ying Zhang, Victor Rafuse and John P Frampton

📅 2026-10-26 📖 1 min 📝 279 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Peripheral nerve regeneration is regulated by structural, biochemical, and electrical cues, yet their combined effects on neurite outgrowth remain unclear. In this study, an aligned fibrous scaffold was developed to evaluate how extracellular matrix (ECM) composition and electrical stimul...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb32c

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

> **Summary:** Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

> **Summary:** Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261101234719&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Emerging light-driven neuromorphic hardware for artificial intelligence

Dimitris Tsoukalas 2026-10-31 1 min 58 words LOW VISION

Summary: Optoelectronic neuromorphic systems have emerged as a promising hardware paradigm for next-generation artificial intelligence, combining the high bandwidth, parallelism, and wavelength selectivity of photonics with the adaptive plasticity of electronic materials. This mini-review provides a focused ...

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Systemic, RPE-directed AAV-tyrosinase therapy restores ocular pigmentation in an OCA1 mouse model

Leah C Byrne 2026-10-31 1 min 71 words LOW VISION

Summary: Oculocutaneous albinism type 1 (OCA1) is a pigmentation disorder caused by biallelic tyrosinase (TYR) mutations. TYR inactivity results in loss of hair, skin, and eye pigment, which is detrimental for ocular function. Hypopigmentation of iris, retinal pigment epithelium (RPE), and choroid results in...

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Systematic review of atropine cost-effectiveness in childhood myopia management

Long Chiau Ming 2026-10-31 1 min 45 words LOW VISION

Summary: CONCLUSION: Studies demonstrate favourable economic value using diverse metrics, including cost per diopter reduction, cost ratios, and cost-utility estimates, rather than uniformly reported ICERs. Greater economic benefits are observed over longer time horizons, largely driven by the avoidance of h...

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Intramedullary Spinal Tuberculosis Mimicking Neoplasm with Ocular Involvement in an Adolescent: A Case Report

👤 Indri Febriani 📅 2026-10-31 📖 1 min 📝 72 words LOW VISION

Summary: Tuberculosis (TB) continues to represent one of the key health problems of high-burden nations. The childhood population is particularly vulnerable to extrapulmonary TB. Spinal intramedullary TB is extremely uncommon and can imitate neoplastic conditions, leading to a late diagnosis with fatal outco...

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Framework-informed analysis of practice: A qualitative study on climate resilience in primary care in Austria

👤 Katharina Brugger 📅 2026-10-31 📖 1 min 📝 59 words LOW VISION

Summary: CONCLUSIONS: PCPs play a vital role in local climate resilience but require tailored, context-specific guidance aligned with PC's integrated, small-scale, multiprofessional nature. Strategic macro-level objectives are required to support low-barrier implementation of resilience measures. Prioritizin...

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Early applications of apple vision pro in clinical medicine and medical education: a scoping review

👤 Chuxu Wang 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: The Apple Vision Pro (AVP), a next-generation spatial computing device, has demonstrated potential applications in multidisciplinary clinical diagnosis, treatment, and medical education, by leveraging its unique technical architecture and performance advantages. In clinical practice, its fundamental...

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A Distributed Vision Transformer-Based Deep Learning Framework for Early Detection and Classification of Lumbar Disc Degeneration

👤 N Chaitanya Kumar 📅 2026-10-31 📖 1 min 📝 68 words LOW VISION

Summary: Lumbar disc degeneration (LDD) is a principal structural cause of chronic low back pain. Magnetic resonance imaging (MRI) is the reference modality for assessing the intervertebral discs, but severity grading is performed manually, is limited by inter-observer variability, and becomes a bottleneck a...

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A Result-Auditable Protocol for Low-Data Classification of Blue-and-White Porcelain Botanical Motifs Using Frozen Chinese-CLIP Features

👤 Chenghao Yu
📅 2026-10-31 📖 1 min 📝 58 words

LOW VISION

Summary: Low-data visual classification is common in cultural heritage documentation, where expert labels are expensive, and image sources are heterogeneous. This protocol provides result-level audit support for classifying five botanical motifs-peony, lotus, chrysanthemum, plum blossom, and bamboo-on blue-a...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel
2026-10-25 1 min 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

Benjamin Ward-Cherrier 2026-10-31 1 min 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

Zhikun Wu 2026-10-30 1 min 58 words TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan
📅 2026-10-30 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

Shouyan Wang 2026-10-30 1 min 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

Aishi Zhou 2026-10-30 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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The sound of manufactured music: Reviewing the role of artificial stimuli in music cognition research.

2024-01-22 1 min 259 words PSYCHOMUSICOLOGY

Summary: Having participants listen and react to musical stimuli is one of music cognition’s foundational methods. Whereas most researchers have used stimuli adapted from existing musical traditions in such work, others have incorporated artificial stimuli (i.e., stimuli generated specifically for research t...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000304

Music-evoked nostalgia and charitable giving: A cross-cultural study in the United States and Mexico.

📅 2024-01-22 📖 1 min 📝 192 words PSYCHOMUSICOLOGY

Summary: Nostalgia, a past-oriented emotion characterized by complex affective responses, is a pervasive and fundamental human experience. Prior research has demonstrated that nostalgia serves various socioemotional functions, such as promoting a sense of belonging, enhancing one’s perception of meaning in l...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000302

Preferred music listening does not affect cognitive inhibition in young and older adults.

📅 2023-10-12 📖 1 min 📝 227 words PSYCHOMUSICOLOGY

Summary: Previous literature has found links between music listening and cognitive performance. Specifically, background music may play a role in modulating cognitive inhibition. However, determining what type of background music affects cognitive inhibition throughout the lifespan has not been studied. The ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000300

Absolute pitch: A literature review of underlying factors, with special regard to music pedagogy.

📅 2023-07-10 📖 1 min 📝 202 words PSYCHOMUSICOLOGY

Summary: Absolute pitch (AP) is a fairly rare and special phenomenon that has relevance for musicology, psychology, genetics, and neuroscience. AP possessors are able to identify the pitch of an isolated sound or to produce that sound without a reference point. The authors' aim is to review the literature on...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000298

Capturing coordination and intentionality in joint musical improvisation.

📅 2023-08-03 📖 1 min 📝 217 words PSYCHOMUSICOLOGY

Summary: Humans collaborate with each other on a wide variety of tasks that are often largely improvised and unscripted. In this study, we investigated the dynamics of coordination in a joint musical improvisation task, what the effect of intentions is on coordination, and how musicians propagate these inten...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000299

Continuous affect responses to a large diverse set of unfamiliar music: Bayesian time-series and cluster analyses.

📅 2023-04-20 📖 1 min 📝 252 words PSYCHOMUSICOLOGY

Summary: Sixty-nine participants made continuous response judgments of perceived arousal and valence while listening to 30-s extracts of 100 unfamiliar pieces within a novel recommender system. Our purpose was to take advantage of the relatively large number of participants and pieces studied (compared with ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000295

Monthly Updates [July]

📅 2026-07-01 📖 3 min 📝 640 words FMHY

Summary: <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   <p>Added FMHY Plugin to <a href="https://www.flowlau...

🔗 **Read full article:**
https://fmhy.net/posts/july-2026

Monday Daily Thread: Project ideas!

👤 /u/ AutoModerator 📅 2026-11-02 📖 1 min 📝 244 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: Project Ideas 💡</h1> Welcome to our weekly Project Ideas thread! Whether you're a newbie looking for a first project or an expert seeking a new challenge, this is the place for you. <h2>How it Works:</h2> Suggest a Project</st...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xlifaf/monday_daily_thread_project_ideas/

OccupySaaS

👤 misswaterfairy 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://occupysaas.com/ Comments URL: https://news.ycombinator.com/item?id=50331402 Points: 131 # Comments: 84

🔗 **Read full article:**
https://occupysaas.com/

Trigger.dev customer environment variables exposed

👤 anurag 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://trigger.dev/blog/security-incident-nov-2026 Comments URL: https://news.ycombinator.com/item?id=50331784 Points: 29 # Comments: 3<...

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https://trigger.dev/blog/security-incident-nov-2026

AI fan wikis riddled with errors are muddying the waters of games culture

👤 rdmuser 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.eurogamer.net/ai-fan-wikis-errors-complicating-games-culture Comments URL: <a href="https://news.ycombinator.com/item?id=50332086">https://news.ycombinator.com/item?id=50332086</...

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https://www.eurogamer.net/ai-fan-wikis-errors-complicating-games-culture

F/OSS Comics #13: BSD Unix, Part 3 – The Beginning of BSD Unix

👤 joonehur 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://fosscomics.com/posts/13.%20BSD%20Unix%20Part%203%20-%20The%20Beginning%20of%20BSD%20Unix/ Comments URL: <a href="https://news.ycombinator.com/item?id=5033216...

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https://fosscomics.com/posts/13.%20BSD%20Unix%20Part%203%20-%20The%20Beginning%20of%20BSD%20Unix/

Type S and M errors as a “rhetorical tool”

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-09-28 📖 17 min 📝 3572 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Update 30/09/2025: I have added a reply by Andrew Gelman below my original blog post. We recently posted a preprint criticizing the idea of Type S and M errors (https://osf.io/2phzb_v1). From our abstract: “While these concepts have been pr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/09/type-s-and-m-errors-as-rhetorical-tool.html

Why we should stop using statistical techniques that have not been adequately vetted by experts in psychology

noreply@blogger.com (Daniel Lakens) 2025-10-29 27 min 5516 words

TWENTY PERCENT STATISTICIAN

Summary: In a recent post on Bluesky, where Richard Morey reflects on a paper he published with Clintin Davis-Stober that points out concerns with the p-curve method (Morey & Davis-Stober, 2025), he writes: <p cla...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/10/why-we-should-stop-using-statistical.html

ERP Decoding for Everyone: Software and Webinar

Steve Luck 2023-06-23 2 min 420 words

ERP BOOT CAMP

Summary: You can access the recording h strong>.
You can access the final PDF of the slides <a href="https://ucdavis.box.com/s/f...

🔗 **Read full article:**
https://erpinfo.org/blog/2023/6/23/decoding-webinar

Important Changes to the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-05 📖 2 min 📝 444 words ERP BOOT CAMP

> **Summary:** We are disappointed to announce that we will not be holding a regular 10-day ERP Boot Camp this summer. We have held Boot Camps nearly every summer since 2007, supported by a series of generous grants from NIMH that allowed us to provide scholarships for all attendees. Unf...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/5/changes-to-the-2024-erp-boot-camp

Standardized Measurement Error now available in BrainVision Analyzer and MNE-Python

👤 Steve Luck 📅 2025-12-19 📖 1 min 📝 253 words ERP BOOT CAMP

> **Summary:** A few years ago, we developed a universal metric of data quality for averaged ERPs called the standardized measurement error (click here for more information or watch our <a href="https://www.youtube.com/watch?v=tEKsx0p53rs" t...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/12/19/sme-now-available-in-bva-and-mne-python

Transcranial Focused Ultrasound: A Transformative Tool for Intracranial Ablation, Drug Delivery, and Neuromodulation

📅 2025-11-11 📖 1 min 📝 136 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Transcranial focused ultrasound (tFUS) is an emerging neuromodulation and therapeutic technology offering noninvasive, submillimeter precision for targeting deep brain structures. Unlike transcranial magnetic stimulation (TMS) and transcranial electric stimulation (tES), which are limited by depth-f...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11241049

Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression

📅 2026-01-16 📖 1 min 📝 196 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Neurodegenerative diseases are characterized by the accumulation of misfolded proteins and widespread disruptions in brain function. Computational modeling has advanced our understanding of these processes, but efforts have traditionally focused on either neuronal dynamics or the biological processe...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11357276

Optogenetics: Pinpoint Light on Precise Neuromodulation

📅 2025-11-06 📖 1 min 📝 137 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Optogenetics has emerged as a pivotal tool in neuroscience, enabling intricate modulation of targeted neurons within the nervous system. Despite its transformative potential, achieving high spatiotemporal resolution in neuromodulation remains a significant challenge, particularly in free-behaving an...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11230544

Emerging Noninvasive Approaches for the Suppression of Pathological Tremor

📅 2025-12-16 📖 1 min 📝 215 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Pathological tremor affects over 40 million people worldwide, significantly impairing daily activities and quality of life. Pharmacological treatments show limited efficacy, with up to 30% discontinuation rates, while surgical interventions like deep brain stimulation achieve significant tremor redu...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11301615

Assistive Trajectory Planning for Lower Limb Exoskeletons: Strategies From Laboratory-Optimized Gait to Environmentally-Adaptive Locomotion Through Multimodal Parameter Awareness

📅 2025-12-30 📖 1 min 📝 182 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Lower limb assistive exoskeletons (LLEs) show great potential to reduce metabolic cost, improve walking performance, and correct abnormal gait patterns. Among their core control architectures, assistive trajectory planning is key to determining system responsiveness and effectiveness under varying l...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11319164

Join the Mailing List

👤 dziura 📅 2025-12-10 📖 1 min 📝 13 words EMBS

Summary: The post Join the Mailing List appeared first on IEEE EMBS.

🔗 **Read full article:**
https://embc.embs.org/2026/#new_tab

EMBS MentorLink: Fostering Careers Together

Deidre Artis 2026-08-18 1 min 14 words EMBS

> **Summary:** The post EMBS MentorLink: Fostering Careers Together appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/embs-mentorlink-fostering-careers-together/

Do neuroimaging data have higher-order interactions?

1 min 16 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Rikkert Hindriks, Fernando Santos, Matteo Neri, Prejaas Tewarie

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006117?dgcid=rss_sd_all

The ViRA-L (Visual Recognition Assessment-Leuven): Development, Validation, and Age Stratified Normative Characterization of a Multi-Category Visual Recognition Battery

👤 Soen, L., Gillebert, C. R., Op de Beeck, H.

📅 2026-11-01 📖 1 min 📝 189 words

BIORXIV NEUROSCIENCE

Summary: Objective: To develop and validate the Visual Recognition Assessment-Leuven (ViRA-L), a recognition-memory battery assessing ten visual categories within a matched task structure, and provide age-stratified normative data. Method: Healthy Dutch-speaking adults completed a visual recognition task for...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.761867v1?rss=1

Infection-related images reduce cortical edge metastability: a specification-curve analysis of a block-design fMRI study

Choi, J., Berjaga-Buisan, T., Garcia-Guzman, E., Sugiura, M., Deco, G.

2026-11-01 1 min

209 words BIORXIV NEUROSCIENCE

Summary: Cues of infection evoke disgust, attentional capture, withdrawal and shifts in social orientation, a set of responses attributed to the behavioral immune system. Responses this varied have to be coordinated, as the effects of inflammation are, so they should be visible in whole-cortex coupling. We p...

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https://www.biorxiv.org/content/10.64898/2026.10.27.762178v1?rss=1

Molecular MRI of Neuroinflammation in Alzheimer's Disease using a Folate-targeted Nanoprobe

Badachhape, A. A., Bhandari, P., Khan, R., Bhavane, R., Menon, R., Parekh, P., Admane, P., Ngan, E., Srivastava, M., Tanifum, E. A., Annapragada, A. V., Ritzel, R. M., Ghaghada, K. B.

2026-11-01 1 min 257 words BIORXIV NEUROSCIENCE

Summary: Background: Neuroinflammation, driven by activated microglia, plays a central role in the pathogenesis of Alzheimer's disease (AD). Non-invasive imaging of neuroinflammation holds promise for advancing our understanding of disease mechanisms and evaluating anti-inflammatory therapeutic strategies. T...

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https://www.biorxiv.org/content/10.64898/2026.10.27.761963v1?rss=1

Community-based RoboGym training improves gross motor function in ambulatory children with cerebral palsy

📅 2026-11-01 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77887-7

Spatial and temporal context effects in visual perception: crowding, serial dependence, and their interaction

📅 2026-11-01 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-76678-4

Editorial: Women in interacting minds and brains II

👤 Anna Ilona Roberts 📅 2026-10-26 📖 1 min 📝 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2001078

Imaging findings and clinical outcomes in neonates with isolated subependymal echogenicity on cranial ultrasound: a retrospective cohort study

Xuehua Zhang 2026-10-28 1 min 255 words FRONTIERS HUMAN NEUROSCIENCE

Summary: ObjectiveTo describe clinical and imaging findings in neonates with isolated subependymal echogenicity on cranial ultrasound and examine associations with recorded mortality and cerebral palsy.MethodsThis retrospective single-center cohort included 317 neonates admitted between January 2015 and Dece...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1945002

Effects of non-invasive brain stimulation combined with exercise training on pain and physical function in knee osteoarthritis: a systematic review and meta-analysis

Zhaoxiang Meng 2026-10-28 1 min 345 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundNon-invasive brain stimulation (NIBS) combined with exercise training (ET) is a promising treatment for knee osteoarthritis (KOA), but its effects on pain and physical function lack systematic evaluation.PurposeTo evaluate the effects of NIBS combined with ET on pain and physical function ...

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-16 1 min 222 words FRONTIERS NEUROROBOTICS

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1925089

Oculomotor rehabilitation after concussion in adults (ORCA): a feasibility study

Tatyana Mollayeva 2026-10-30 1 min 313 words FRONTIERS NEUROSCIENCE

Summary: BackgroundOculomotor deficits are reported in up to 85% of adults after concussion and may contribute to prolonged symptoms and delayed return to daily activities. Oculomotor rehabilitation has been successfully implemented in younger age groups, however evidence in adults is lacking. Virtual realit...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1898653

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Emerging light-driven neuromorphic hardware for artificial intelligence

Dimitris Tsoukalas 2026-10-31 1 min 58 words LOW VISION

Summary: Optoelectronic neuromorphic systems have emerged as a promising hardware paradigm for next-generation artificial intelligence, combining the high bandwidth, parallelism, and wavelength selectivity of photonics with the adaptive plasticity of electronic materials. This mini-review provides a focused ...

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Systemic, RPE-directed AAV-tyrosinase therapy restores ocular pigmentation in an OCA1 mouse model

Leah C Byrne 2026-10-31 1 min 71 words LOW VISION

Summary: Oculocutaneous albinism type 1 (OCA1) is a pigmentation disorder caused by biallelic tyrosinase (TYR) mutations. TYR inactivity results in loss of hair, skin, and eye pigment, which is detrimental for ocular function. Hypopigmentation of iris, retinal pigment epithelium (RPE), and choroid results in...

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Systematic review of atropine cost-effectiveness in childhood myopia management

👤 Long Chiau Ming 📅 2026-10-31 📖 1 min 📝 45 words LOW VISION

Summary: CONCLUSION: Studies demonstrate favourable economic value using diverse metrics, including cost per diopter reduction, cost ratios, and cost-utility estimates, rather than uniformly reported ICERs. Greater economic benefits are observed over longer time horizons, largely driven by the avoidance of h...

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Intramedullary Spinal Tuberculosis Mimicking Neoplasm with Ocular Involvement in an Adolescent: A Case Report

👤 Indri Febriani 📅 2026-10-31 📖 1 min 📝 72 words LOW VISION

Summary: Tuberculosis (TB) continues to represent one of the key health problems of high-burden nations. The childhood population is particularly vulnerable to extrapulmonary TB. Spinal intramedullary TB is extremely uncommon and can imitate neoplastic conditions, leading to a late diagnosis with fatal outco...

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Framework-informed analysis of practice: A qualitative study on climate resilience in primary care in Austria

Katharina Brugger 2026-10-31 1 min 59 words LOW VISION

Summary: CONCLUSIONS: PCPs play a vital role in local climate resilience but require tailored, context-specific guidance aligned with PC's integrated, small-scale, multiprofessional nature. Strategic macro-level objectives are required to support low-barrier implementation of resilience measures. Prioritizin...

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Early applications of apple vision pro in clinical medicine and medical education: a scoping review

Chuxu Wang 2026-10-31 1 min 58 words LOW VISION

Summary: The Apple Vision Pro (AVP), a next-generation spatial computing device, has demonstrated potential applications in multidisciplinary clinical diagnosis, treatment, and medical education, by leveraging its unique technical architecture and performance advantages. In clinical practice, its fundamental...

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A Distributed Vision Transformer-Based Deep Learning Framework for Early Detection and Classification of Lumbar Disc Degeneration

👤 N Chaitanya Kumar
📅 2026-10-31 📖 1 min 📝 68 words

LOW VISION

Summary: Lumbar disc degeneration (LDD) is a principal structural cause of chronic low back pain. Magnetic resonance imaging (MRI) is the reference modality for assessing the intervertebral discs, but severity grading is performed manually, is limited by inter-observer variability, and becomes a bottleneck a...

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A Result-Auditable Protocol for Low-Data Classification of Blue-and-White Porcelain Botanical Motifs Using Frozen Chinese-CLIP Features

👤 Chenghao Yu
📅 2026-10-31 📖 1 min 📝 58 words

LOW VISION

Summary: Low-data visual classification is common in cultural heritage documentation, where expert labels are expensive, and image sources are heterogeneous. This protocol provides result-level audit support for classifying five botanical motifs-peony, lotus, chrysanthemum, plum blossom, and bamboo-on blue-a...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

👤 Alan J Emanuel
📅 2026-10-25 📖 1 min 📝 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

Benjamin Ward-Cherrier 2026-10-31 1 min 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

Zhikun Wu 2026-10-30 1 min 58 words TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan
📅 2026-10-30 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

Chuhuai Wang
2026-10-31 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao | 2026-11-01 | 1 min | 61 words | TDCS TACS TRNS

> **Summary:** The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay | 2026-10-29 | 1 min | 62 words | FNIRS

> **Summary:** Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

2026-07-30 1 min 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001111

Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001093

The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001085

Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001087

Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001096

litellm's bad release was up for under 2 hours, so our PyPI installs now wait a week

👤 /u/ iApexxx 📅 2026-11-01 📖 1 min 📝 178 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> litellm 1.107.0 was on PyPI for 1 h 43 min on 21 October, and its payload sat in a <code>.pth</code> file, so it ran whenever Python started in that environment, no import needed. lightning 2.6.2/2.6.3 lasted 42 minutes back in April. Both were gone long befo...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xlfsel/litellms_bad_release_was_up_for_under_2_hours_so/

Landline phones will switch off across Britland in 100 days, no extensions

👤 edward 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: <a href="https://www.theregister.com/networks/2026/11/01/landline-phones-will-switch-off-across-britland-in-100-days-no-extensions-openreach/5306673">https://www.theregister.com/networks/2026/11/01/landline-phones-will-switch-off-across-britland-in-100-days-no-extensions-openreach/53...

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John Leeson (the voice of K9) has died

👤 slyrus 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.doctorwhonews.net/2026/11/john_leeson_1943_2026.html Comments URL: https://news.ycombinator.com/item?id=50329707 Points...

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AI delivers 'zero financial payoff' to most companies, Apollo economist says

👤 throwoutway 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://techcrunch.com/2026/10/25/ai-delivers-zero-financial-payoff-to-most-companies-according-to-apollo-economist/ Comments URL: <a href="https:...

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https://techcrunch.com/2026/10/25/ai-delivers-zero-financial-payoff-to-most-companies-according-to-apollo-economist/

U.S. workers took home their smallest share of capital since at least 1947

👤 qlte 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://fortune.com/article/why-did-us-workers-just-take-home-smallest-share-of-capital-since-1947-lowest-ever-recorded/ Comments URL: <a href...

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Automatic computational reproducibility checks with Metacheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-10-30 📖 19 min 📝 3854 words

TWENTY PERCENT STATISTICIAN

> **Summary:** When authors share their data and analysis code, the idea is that anyone should be able to run that code on that data and get the results reported in the paper. But sharing data and code in a way that results are reproducible is difficult, and researchers have often not received tr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/automatic-computational-reproducibility.html

Registration is now full for the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-16 📖 1 min 📝 106 words ERP BOOT CAMP

> **Summary:** The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

Deidre Artis 2026-09-29 1 min 14 words EMBS

Summary: The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Stage-specific exposure to an activity-permissive medium enhances neuronal activity in oligodendrocyte-enriched cortical organoids

1 min 34 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Clara Chung, Michelle J. Kim, Grace Field, Konstantina Pilarinos, Elizabeth K. Kharitonova, Natalie Baker Campbell, Marie Shi, Christopher V. Gabel, Joseph L. Orofino, Ella Zeldich

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004634?dgcid=rss_sd_all

Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zuhal Abasiyanik, Lousin Moumdjian, Natalia Skorupa, Daphne Kos, Bart Van Wijmeersch, Miguel D’Haeseleer, Inge Zijdewind, Peter Feys, Felipe Balistieri Santinelli

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006949?dgcid=rss_sd_all

Neural dynamics of auditory attention: Interactions between top-down and bottom-up processes

📖 1 min 📝 12 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): S. Graceffo, M. Elhilali

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006002?dgcid=rss_sd_all

Neurovascular coupling in simultaneous EEG-fNIRS studies: A scoping review of analytical frameworks and clinical applications

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Yu Wu, Rajamanickam Yuvaraj, Stéphane Perrey, Hasan Ayaz, Wei-Peng Teo

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006130?dgcid=rss_sd_all

Phenotypic screening in Parkinson's iPSC-dopamine neurons identifies the FASN-SREBP1 axis as a potential therapeutic target

👤 Caiazza, M. C., Brandes, J., Barrow, E. R., Marquez-Gomez, R., Brown, D., Jenkins, B., Chughtai, U., McGuinness, W., Humble, S. W., Sharifi, P., Hoffmann, J. L., Trigano, M., Koulman, A., Gurrell, I., Davis, A., Smith, D., Ryan, B., Wade-Martins, R.

📅 2026-11-01 📖 1 min 📝 250 words BIORXIV NEUROSCIENCE

Summary: Parkinson's disease (PD) remains highly challenging to model in vitro, particularly when seeking robust functional phenotypes for target discovery. Patient-derived induced pluripotent stem cell dopaminergic neurons (iPSC-DANs) offer a promising platform to address this challenge. Because calcium dys...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.761977v1?rss=1

Behavioral deficits in Caenorhabditis elegans raised for six generations under Mars-analog gravity and hypomagnetic fields

📅 2026-11-01 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78119-8

C1ql2 gates stress vulnerability by regulating microglia-mediated synaptic pruning in the BLA-vHPC circuit

👤 Chun-Yan WangWei-Zhu LiuShuang-Mei LiaoJing NiuHan-Qing PanWen-Jie YouKe ZhaoBao MaMing-Fen LuJia-Xin ZouPing HuWen-Hua ZhangBing-Xing Panahttps://ror.org/042v6xz23Jiangxi Institute of Respiratory Diseases and Department of Pathology, The First Affiliated Hospital, School of Basic Medical Sciences and Institute of Biomedical Innovation, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinabhttps://ror.org/042v6xz23Jiangxi Provincial Key Laboratory of Brain Science and Brain Health, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinachttps://ror.org/042v6xz23Postdoctoral Research Station, The First Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinadhttps://ror.org/042v6xz23Institute of Translational Medicine, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinaehttps://ror.org/042v6xz23Department of Neurology, The Second Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, China

📅 2026-10-21 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceStress-related emotional disorders involve maladaptive synaptic remodeling in the basolateral amygdala (BLA), which is highly heterogeneous across neurons projecting to distinct downstream targets....

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2600854123?af=R

BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

Tingwei Quan 2026-10-07 1 min 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1925112

Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

Ansel Y. Rodríguez-González 2026-10-07 1 min 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1919962

Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

Bocheng Wang 2026-10-07 1 min 217 words FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1869687

Mechanonociceptive tactile feedback enhances prosthetic grasp control during repeated human-in-the-loop interaction

Elif Hocaoglu 2026-10-15 1 min 215 words FRONTIERS NEUROROBOTICS

Summary: Upper-limb prosthetic systems continue to face a fundamental challenge in providing sensory feedback that remains functionally effective during repeated interaction. This study investigates innocuous painful feedback (IPF), a controlled mechanonociceptive tactile modality intended to provide salient...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1910158

Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

Peng Han 2026-10-22 1 min 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1934032

MS-3D-CNN with quantum-inspired chaotic hybrid adaptive wolf-particle swarm optimization for tactical pattern classification in basketball from player tracking data

Haoyan Liu 2026-10-30 1 min 229 words FRONTIERS NEUROROBOTICS

Summary: IntroductionAI sport analysis methods have been considerably improved by developments in knowledge of tactically information and its codification. Nevertheless, methods for capturing multi-levels spatio-temporal information for complex team-sport behaviors are still limited. In opposed to motion rec...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1910261

Cross-subject EEG driving fatigue detection: early personal spectral reference and causal long-sequence residual encoding

Lanlan Lv 2026-10-30 1 min 265 words FRONTIERS NEUROSCIENCE

Summary: BackgroundIn cross-subject electroencephalography (EEG)-based driving-fatigue detection, most existing decoders are designed around second-scale short windows, whereas fatigue is a slow drift; short-window spectral features provide little direct evidence of that slow drift, limiting fatigue decision...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1997832

A progenitor with immature Bergmann glia-like characteristics in the developing cerebellar cortex

Kazuhiko Sawada 2026-10-30 1 min 203 words FRONTIERS NEUROSCIENCE

Summary: The cerebellum is an ancient brain component, conserved across mammalian species, that has classically played roles in regulating muscle tone, limb coordination, and equilibrium, as well as in non-motor functions, such as language, cognition, and emotion. Recently, we reported Nestin-expressing cell...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1963374

Cerebral blood flow asymmetry in motor-related regions is associated with 30-day upper-limb motor recovery after acute ischemic stroke

Xin Wang 2026-10-30 1 min 337 words FRONTIERS NEUROSCIENCE

Summary: BackgroundUpper-limb motor impairment is a leading cause of long-term disability after acute ischemic stroke (AIS), yet early prognostication relies mainly on clinical scales. Cerebral blood flow (CBF) asymmetry measured by arterial spin labeling (ASL) perfusion MRI may indicate recovery potential. ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1955245

Glial handling of alpha-synuclein seeds: uptake, degradation and redistribution in pathological propagation

R. M. Meade 2026-10-30 1 min 202 words FRONTIERS NEUROSCIENCE

Summary: α-synuclein (αS) aggregation and propagation are central features of Parkinson's disease pathology. Preformed fibril (PFF) models have shown that extracellular αS aggregates can enter neurons and seed aggregation of endogenous αS. Within the brain, these seeds also encounter microglia and astrocytes...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.2017355

Case Report: A novel hemizygous nonsense variant in PPP1R3F gene associated with developmental delay, intellectual disability and autism spectrum disorder

Jingsi Luo 2026-10-30 1 min 328 words FRONTIERS NEUROSCIENCE

Summary: BackgroundPPP1R3F deficiency syndrome is an extremely rare, recently identified X-linked recessive neurodevelopmental disorder. It is caused by hemizygous variants in the PPP1R3F gene, which encodes a glycogen-targeting subunit of the brain-specific protein phosphatase 1 (PP1). Dysfunction of PPP1R3...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1998834

Ordered onset of regional desynchronization and a transition in movement-class discriminability during cue-paced motor imagery

Seho Lee 2026-10-21 1 min 298 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Motor imagery recruits a distributed network whose components and large-scale organization are well characterized. The temporal order in which those components are engaged is not: the latency at which each region begins to desynchronize within a trial has not been compared statistically i...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb161

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

Robert Oostenveld 2026-02-12 1 min 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Emerging light-driven neuromorphic hardware for artificial intelligence

👤 Dimitris Tsoukalas 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: Optoelectronic neuromorphic systems have emerged as a promising hardware paradigm for next-generation artificial intelligence, combining the high bandwidth, parallelism, and wavelength selectivity of photonics with the adaptive plasticity of electronic materials. This mini-review provides a focused ...

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Systemic, RPE-directed AAV-tyrosinase therapy restores ocular pigmentation in an OCA1 mouse model

👤 Leah C Byrne 📅 2026-10-31 📖 1 min 📝 71 words LOW VISION

Summary: Oculocutaneous albinism type 1 (OCA1) is a pigmentation disorder caused by biallelic tyrosinase (TYR) mutations. TYR inactivity results in loss of hair, skin, and eye pigment, which is detrimental for ocular function. Hypopigmentation of iris, retinal pigment epithelium (RPE), and choroid results in...

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Systematic review of atropine cost-effectiveness in childhood myopia management

👤 Long Chiau Ming 📅 2026-10-31 📖 1 min 📝 45 words LOW VISION

Summary: CONCLUSION: Studies demonstrate favourable economic value using diverse metrics, including cost per diopter reduction, cost ratios, and cost-utility estimates, rather than uniformly reported ICERs. Greater economic benefits are observed over longer time horizons, largely driven by the avoidance of h...

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Intramedullary Spinal Tuberculosis Mimicking Neoplasm with Ocular Involvement in an Adolescent: A Case Report

👤 Indri Febriani 📅 2026-10-31 📖 1 min 📝 72 words LOW VISION

Summary: Tuberculosis (TB) continues to represent one of the key health problems of high-burden nations. The childhood population is particularly vulnerable to extrapulmonary TB. Spinal intramedullary TB is extremely uncommon and can imitate neoplastic conditions, leading to a late diagnosis with fatal outco...

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Framework-informed analysis of practice: A qualitative study on climate resilience in primary care in Austria

👤 Katharina Brugger 📅 2026-10-31 📖 1 min 📝 59 words LOW VISION

Summary: CONCLUSIONS: PCPs play a vital role in local climate resilience but require tailored, context-specific guidance aligned with PC's integrated, small-scale, multiprofessional nature. Strategic macro-level objectives are required to support low-barrier implementation of resilience measures. Prioritizin...

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Early applications of apple vision pro in clinical medicine and medical education: a scoping review

👤 Chuxu Wang 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: The Apple Vision Pro (AVP), a next-generation spatial computing device, has demonstrated potential applications in multidisciplinary clinical diagnosis, treatment, and medical education, by leveraging its unique technical architecture and performance advantages. In clinical practice, its fundamental...

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A Distributed Vision Transformer-Based Deep Learning Framework for Early Detection and Classification of Lumbar Disc Degeneration

👤 N Chaitanya Kumar
📅 2026-10-31 📖 1 min 📝 68 words

LOW VISION

Summary: Lumbar disc degeneration (LDD) is a principal structural cause of chronic low back pain. Magnetic resonance imaging (MRI) is the reference modality for assessing the intervertebral discs, but severity grading is performed manually, is limited by inter-observer variability, and becomes a bottleneck a...

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A Result-Auditable Protocol for Low-Data Classification of Blue-and-White Porcelain Botanical Motifs Using Frozen Chinese-CLIP Features

👤 Chenghao Yu
📅 2026-10-31 📖 1 min 📝 58 words

LOW VISION

Summary: Low-data visual classification is common in cultural heritage documentation, where expert labels are expensive, and image sources are heterogeneous. This protocol provides result-level audit support for classifying five botanical motifs-peony, lotus, chrysanthemum, plum blossom, and bamboo-on blue-a...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

👤 Alan J Emanuel
📅 2026-10-25 📖 1 min 📝 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

👤 Zhikun Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan
📅 2026-10-30 📖 1 min 📝 33 words

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Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Psychomusicology: A resounding closing cadence.

2024-01-22 1 min 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000403

The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual’s cognition in solving a task. However, very little is known about how an individual’s conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000400

Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

📅 2026-09-10 📖 1 min 📝 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

> **Summary:** Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

[Blog Post] The State of Python Profiling in 2026

👤 /u/P403n1x87 📅 2026-11-01 📖 1 min 📝 119 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Python profiling has changed more in the last two years than in the decade before. With CPython 3.15 introducing a sampling profiler directly into the standard library (Tachyon) and free-threading going mainstream, the landscape looks vibrant as never before. <p...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xlcdgb/blog_post_the_state_of_python_profiling_in_2026/

Show HN: HackerBooks – HackerTube, but for books mentioned on HN

👤 roansh 📅 2026-11-01 📖 1 min 📝 115 words HACKER NEWS

> **Summary:** Saw Karpathy's HackerTube and wanted the same thing for books. So I had Claude build one (Opus 5.5). It's ~16.5k books that at least two different HN commenters brought up, from 2007 till now. Each book links back to the comments that mentioned it. Data is downloadable from the about page. To...

🔗 **Read full article:**
https://hackerbooks.org

Seoul Viosys begins production of rare-earth-free white light emitter

👤 gmays 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.semiconductor-today.com/news_items/2026/oct/seoul-291026.shtml Comments URL: <a href="https://news.ycombinator.com/item?id=50327058">https://news.ycombinator.com/item?id=503270...

🔗 **Read full article:**
https://www.semiconductor-today.com/news_items/2026/oct/seoul-291026.shtml

Frontier Airlines Likely Breached

👤 999900000999 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.reddit.com/r/frontierairlines/comments/1xl20e8/psa_email_scam/ Comments URL: <a href="https://news.ycombinator.com/item?id=50327266">https://news.ycombinator.com/item?id=503272...

🔗 **Read full article:**
https://www.reddit.com/r/frontierairlines/comments/1xl20e8/psa_email_scam/

Patrick Stewart taught us what good men do with power

👤 rhaynes 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.vox.com/politics/508526/patrick-stewart-jean-luc-picard-star-trek-liberalism Comments URL: <a href="https://news.ycombinator.com/item?id=50327650">https://news.yc...

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https://www.vox.com/politics/508526/patrick-stewart-jean-luc-picard-star-trek-liberalism

Mathematics has changed. Let's act like it

👤 smilelamp 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://terrytao.wordpress.com/2026/11/01/mathematics-has-changed-lets-act-like-it/ Comments URL: <a href="https://news.ycombinator.com/item?id=50328251">https://news.ycombinator....

🔗 **Read full article:**
https://terrytao.wordpress.com/2026/11/01/mathematics-has-changed-lets-act-like-it/

US Jobs Pickup May Keep Fed on Track to Hike Rates Again

👤 frank_clover 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.bloomberg.com/news/articles/2026-10-31/us-jobs-pickup-may-keep-fed-on-track-to-hike-rates-again Comments URL: <a href="https://news.ycombinator...

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https://www.bloomberg.com/news/articles/2026-10-31/us-jobs-pickup-may-keep-fed-on-track-to-hike-rates-again

Nurses Sound Alarm as 'Uber for Nursing' Apps Push to Deregulate Healthcare

👤 ravetcofx 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://nurse.org/news/uber-for-nursing-gig-apps-report/ Comments URL: https://news.ycombinator.com/item?id=50328487 Points: 83 # Co...

🔗 **Read full article:**
https://nurse.org/news/uber-for-nursing-gig-apps-report/

It wasn't AI that flipped Proton's charge

👤 Raed667 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.raed.dev/posts/proton-loser-thinking/ Comments URL: https://news.ycombinator.com/item?id=50328488 Points: 21 # Comments: 2</p...

🔗 **Read full article:**
https://blog.raed.dev/posts/proton-loser-thinking/

Ask HN: Which TV do you recommend for privacy?

👤 LecroJS 📅 2026-11-01 📖 1 min 📝 110 words HACKER NEWS

> **Summary:** I haven’t owned a TV in years and would like one that respects my privacy. Are there any TV’s which do this well? I am looking for something in the 60”to 80”range and would like a 4k display with at least 60hz display. I don’t mind paying a premium. I have heard good things about Apple TV as a ...

🔗 **Read full article:**
https://news.ycombinator.com/item?id=50328896

Charitable giving with earned and windfall money: behavioral and ERP responses to donation-risk information

📖 1 min 📝 17 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Jing Wang, Xiaoya Li, Siqi Jian, Peng Li

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004750?dgcid=rss_sd_all

Transcutaneous auricular vagus nerve stimulation attenuates post-stroke dysphagia by modulating microglial polarization via the endothelial VWF/PI3K-Akt axis

📖 1 min 📝 17 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Fan Yu, Yi Feng, Yun Miao, Guangxu Xu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004671?dgcid=rss_sd_all

TDP-43 dysfunction destabilizes glial homeostatic identity in ALS/FTD

📖 1 min 📝 14 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Anna E Dale, Rita Sattler

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004804?dgcid=rss_sd_all

Cannabidiol’s prosexual effects in sexually active and sexually inhibited male rats

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Gabriela Rodríguez-Manzo, Ana Canseco-Alba, Jorge Rojas-Hernández

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006998?dgcid=rss_sd_all

Machine learning of connectome-wide functional connectivity in spinal muscular atrophy

📖 1 min 📝 36 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Yufen Li, Huasong Cai, Mengyu Zhang, Mengzhen Yan, Shasha Lan, Huirong Nie, Wanqing Shen, Cui Yan, Boyan Xu, Yujian Liang, Wen Tang, Yijuan Li, Pei Xiang, Yingqian Chen

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007037?dgcid=rss_sd_all

Spectral-spatial modeling of neural couplings in functional connectivity of motor networks

📖 1 min 📝 25 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Hamidreza Abbaspour, Melissa H. Wade, Christopher P. Hurt, Christopher L. Gonzalez, Harrison C. Walker, Zachary T. Irwin

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006294?dgcid=rss_sd_all

Human iEEG recordings during audiovisual speech violations support and extend predictive routing models

👤 Collado, J. V., Nejat, H., Brang, D., Wasade, V. S., Stacey, W. C., Bastos, A. M. 📅 2026-11-01

📖 1 min 📝 243 words BIORXIV NEUROSCIENCE

> **Summary:** Rather than constantly reconstructing the world solely through sensory input, predictive processing theories posit that the brain utilizes predictions maintained as internal representations to guide sensory processing. An emerging model, predictive routing, proposes that an interplay between neurona...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.760243v1?rss=1

The human-specific gene FRMPD2B couples postnatal experience to the expansion of cortical connectivity

Christopoulou, E., Zamith, J., Bonadonna, P., Rivero Romano, M., Vaudin, M., Nguyen, W., Brunet, C., Blugeon, C., Lemoine, S., Giudicelli, F., Baffet, A., Charrier, C.

2026-11-01 1 min 149 words BIORXIV NEUROSCIENCE

Summary: The human neocortex is characterized by its uniquely prolonged postnatal developmental period during which experience sculpts neuronal circuits, a process essential to human cognitive abilities. Yet whether human-specific genetic innovations have shaped the activity-dependent mechanisms regulating c...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763086v1?rss=1

Superior Frontal Gyrus Cortical Thickness Mediates the Association Between Openness to Experience and Theory of Mind: Evidence from the Human Connectome Project

Akar, B., Atasever, U.

2026-11-01 1 min 256 words BIORXIV NEUROSCIENCE

Summary: Background: Theory of mind (ToM), the capacity to infer the mental states of others, varies considerably across healthy adults. Openness to experience has been linked both to social-cognitive abilities and to thinner prefrontal cortex, but whether cortical structure statistically accounts for the as...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.31.763281v1?rss=1

A Unified World Model in Human Posterior Parietal Cortex

Bougou, V., Pramod, R., Rosario, E. R., Pejsa, K., Bari, A., Tenenbaum, J. B., Kanwisher, N., Andersen, R. A.

2026-11-01 1 min 124 words BIORXIV NEUROSCIENCE

Summary: How does the brain understand the physical world, predict future events, and plan actions? Are these problems solved by a unified world model, as hypothesized by cognitive scientists and artificial intelligence researchers? Posterior parietal cortex has been implicated in all these functions, but wh...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.30.762936v1?rss=1

Cortical long-range dynamics track excitation-inhibition balance across cognitive states and organizational principles

Bisi, M., Ferrari, A., Degutis, J. K., Tambalo, S., Renton, A. I., Lombardo, M. V., Jovicich, J., Van De Ville, D., Saviola, F.

2026-11-01 1 min 160 words BIORXIV NEUROSCIENCE

Summary: A fundamental question in systems neuroscience is how the brain's excitation-inhibition (E/I) balance shapes the temporal structure of neural activity and whether we can non-invasively recover this relationship from neural scaling properties. We show that MRS-derived E/I balance is expressed in long...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.30.763069v1?rss=1

Xenogeneic acellular nerve grafts laden with Schwann cell-like cells repair monkey long-gap facial nerve defects

📅 2026-11-01 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41536-026-00518-2

Serial multi-time-point assessments of well-defined somatosensory domains and their associations with limb spasticity after ischemic stroke

📅 2026-11-01 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78189-8

Delineating neural contributions to electroencephalogram-based speech decoding

Motoshige Sato, Eri Hatakeyama, Masakazu Inoue, Yasuo Kabe, Sensho Nobe, Akito Yoshida, Atsushi Yamamoto, Yuya Kita, Mayumi Shimizu, Kenichi Tomeoka, Michael X Cohen and Shuntaro Sasai

2026-10-30 1 min 249 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Speech brain–computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

Read full article:
https://iopscience.iop.org/article/10.1088/1741-2552/aeb68d

Diff-ADN: a diffusion-guided artifact denoising network with deterministic residual refinement for EEG

Mudasir Jamil, Muhammad Zulkifal Aziz, Binwen Huang and Xiaojun Yu

2026-10-30 1 min 241 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Physiological artifacts degrade electroencephalographic (EEG) recordings and can affect downstream brain–computer interface (BCI) analysis. This study develops a compact single-channel framework for ocular, muscular, cardiac, and mixed-artifact removal. Approach. The Diffusion-Guided Arti...

Read full article:
https://iopscience.iop.org/article/10.1088/1741-2552/aeab36

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Emerging light-driven neuromorphic hardware for artificial intelligence

Dimitris Tsoukalas 2026-10-31 1 min 58 words LOW VISION

Summary: Optoelectronic neuromorphic systems have emerged as a promising hardware paradigm for next-generation artificial intelligence, combining the high bandwidth, parallelism, and wavelength selectivity of photonics with the adaptive plasticity of electronic materials. This mini-review provides a focused ...

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Systemic, RPE-directed AAV-tyrosinase therapy restores ocular pigmentation in an OCA1 mouse model

Leah C Byrne 2026-10-31 1 min 71 words LOW VISION

Summary: Oculocutaneous albinism type 1 (OCA1) is a pigmentation disorder caused by biallelic tyrosinase (TYR) mutations. TYR inactivity results in loss of hair, skin, and eye pigment, which is detrimental for ocular function. Hypopigmentation of iris, retinal pigment epithelium (RPE), and choroid results in...

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Systematic review of atropine cost-effectiveness in childhood myopia management

👤 Long Chiau Ming 📅 2026-10-31 📖 1 min 📝 45 words LOW VISION

Summary: CONCLUSION: Studies demonstrate favourable economic value using diverse metrics, including cost per diopter reduction, cost ratios, and cost-utility estimates, rather than uniformly reported ICERs. Greater economic benefits are observed over longer time horizons, largely driven by the avoidance of h...

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Intramedullary Spinal Tuberculosis Mimicking Neoplasm with Ocular Involvement in an Adolescent: A Case Report

👤 Indri Febriani 📅 2026-10-31 📖 1 min 📝 72 words LOW VISION

Summary: Tuberculosis (TB) continues to represent one of the key health problems of high-burden nations. The childhood population is particularly vulnerable to extrapulmonary TB. Spinal intramedullary TB is extremely uncommon and can imitate neoplastic conditions, leading to a late diagnosis with fatal outco...

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Framework-informed analysis of practice: A qualitative study on climate resilience in primary care in Austria

Katharina Brugger 2026-10-31 1 min 59 words LOW VISION

Summary: CONCLUSIONS: PCPs play a vital role in local climate resilience but require tailored, context-specific guidance aligned with PC's integrated, small-scale, multiprofessional nature. Strategic macro-level objectives are required to support low-barrier implementation of resilience measures. Prioritizin...

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Early applications of apple vision pro in clinical medicine and medical education: a scoping review

Chuxu Wang 2026-10-31 1 min 58 words LOW VISION

Summary: The Apple Vision Pro (AVP), a next-generation spatial computing device, has demonstrated potential applications in multidisciplinary clinical diagnosis, treatment, and medical education, by leveraging its unique technical architecture and performance advantages. In clinical practice, its fundamental...

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A Distributed Vision Transformer-Based Deep Learning Framework for Early Detection and Classification of Lumbar Disc Degeneration

👤 N Chaitanya Kumar
📅 2026-10-31 📖 1 min 📝 68 words

LOW VISION

Summary: Lumbar disc degeneration (LDD) is a principal structural cause of chronic low back pain. Magnetic resonance imaging (MRI) is the reference modality for assessing the intervertebral discs, but severity grading is performed manually, is limited by inter-observer variability, and becomes a bottleneck a...

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A Result-Auditable Protocol for Low-Data Classification of Blue-and-White Porcelain Botanical Motifs Using Frozen Chinese-CLIP Features

👤 Chenghao Yu
📅 2026-10-31 📖 1 min 📝 58 words

LOW VISION

Summary: Low-data visual classification is common in cultural heritage documentation, where expert labels are expensive, and image sources are heterogeneous. This protocol provides result-level audit support for classifying five botanical motifs-peony, lotus, chrysanthemum, plum blossom, and bamboo-on blue-a...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

👤 Alan J Emanuel
📅 2026-10-25 📖 1 min 📝 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

👤 Zhikun Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan
📅 2026-10-30 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

👤 Blandina T Mmbaga 📅 2026-10-31 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

👤 Matias Valdenegro-Toro 📅 2026-10-31 📖 1 min 📝 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

2026-07-09 1 min 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000661

A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000660

Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 885 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/april-2026

Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 892 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/may-2026

Monthly Updates [November]

📅 2026-11-01 📖 5 min 📝 1032 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

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https://fmhy.net/posts/nov-2026

Domain hacks with unusual Unicode characters (2018)

👤 n_plus_1_acc 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://xn--69f31l4t57c0mag4b613h.xn--7uh4898msjaso/□□□/ Comments URL: https://news.ycombinator.com/item?id=50324371 Points: 34 # Co...

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Restrooms along the Norwegian Scenic Routes

👤 giuliomagnifico 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.visitnorway.com/plan-your-trip/travel-tips-a-z/norwegian-scenic-routes/worlds-best-toilets/ Comments URL: <a href="https://news.ycombinator.com/ite...

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Show HN: OmniDiff – Opensource, super fast, content aware Git difftool, in Rust

👤 FabCH 📅 2026-11-01 📖 1 min 📝 163 words HACKER NEWS

> **Summary:** I have been researching what the ultimate diff tool would look like, based on actual diffs in open source repositories. OmniDiff can diff: code (25 languages), config, images, PDFs, archives, cursors, fonts... It's fast. It can diff C++ files of tens of thousands of lines in under 100ms, on a...

🔗 **Read full article:**
https://github.com/ivankovic/omnidiff

The "Moon Trees"

👤 momentmaker 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://nssdc.gsfc.nasa.gov/planetary/lunar/moon_tree.html Comments URL: https://news.ycombinator.com/item?id=50325411 Points: 20 ...

🔗 **Read full article:**
https://nssdc.gsfc.nasa.gov/planetary/lunar/moon_tree.html

Proving and Improving the Platonic Representation Hypothesis with JEPAs

👤 Anon84 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://arxiv.org/abs/2610.39302 Comments URL: https://news.ycombinator.com/item?id=50325454 Points: 38 # Comments: 8

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AI agent swarms in Minecraft fail to build civilization, drown in bureaucracy

👤 entryleveljobsc 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: <a href="https://www.tomshardware.com/tech-industry/artificial-intelligence/ai-agent-swarms-unleashed-in-minecraft-to-build-a-thriving-civilization-50-agent-teams-fail-after-falling-into-paralyzing-bureaucracy">https://www.tomshardware.com/tech-industry/artificial-intelligence/ai-age...

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Novellas with Groff

👤 dverlaeckt80 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://movq.de/blog/postings/2026-10-31/0/POSTING-en.html Comments URL: https://news.ycombinator.com/item?id=50325694 Points: 29 ...

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Lenovo's 'encrypted' fingerprint USB is just an unencrypted SD card in a case

👤 entryleveljobsc 📅 2026-11-01 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.reddit.com/r/technology/comments/1xl2pr3/lenovos_encrypted_fingerprint_usb_drive_is_just/ Comments URL: <a href="https://news.ycombinator.com/item?id...

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Police chief unplugs Flock cameras after company ignores contract cancellation

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> **Summary:** <p>Article URL: <a href="https://www.tomshardware.com/tech-industry/big-tech/police-chief-physically-unplugs-flock-cameras-after-the-company-ignores-contract-cancellation-city-takes-matters-into-its-own-hands-amid-privacy-controversies">https://www.tomshardware.com/tech-industry/big-tech/police-chie...

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New software package: ERPLAB Studio

👤 Steve Luck 📅 2024-06-12 📖 2 min 📝 449 words ERP BOOT CAMP

> **Summary:** <p class="">We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it’s a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/11/erplab-studio

A Hierarchical-Dynamic integration model for the formation of interpersonal neural synchrony during spoken communication

📖 1 min 📝 13 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yating Bai, Qingfang Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004567?dgcid=rss_sd_all

Automated multiscale high-content imaging connects plaque population heterogeneity to localized synapse loss

👤 Polino, C., Port, R., Hofmann, M., Smith, S., Wang, X., Klein, R., Marino, M. 📅 2026-10-31 📖 1 min

📝 159 words BIORXIV NEUROSCIENCE

> **Summary:** Synapse loss is a major correlate of cognitive decline in neurodegenerative disease, yet its relationship to localized pathology remains difficult to study at scale. Conventional confocal imaging approaches are limited by low throughput, selection bias, and restricted sampling of pathological featur...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761378v1?rss=1

The Canadian Pediatric Imaging Platform (C-PIP) Reference Sample: Neuroimaging Protocol and Cross-site Validation for Harmonized Multimodal Pediatric Imaging

👤 Singh, M., Parimoo, S., Enguix, V., Cortese, F., Camacho, M., Spinney, S., Lynch, S. J., Carlson, H. L., Leijser, L. M., Tardif, C. L., Leppert, I. R., Pike, G. B., Kadalie, E., Near, J., Kumaragamage, C., Chitroda, R., Lauzon, M. L., Manning, K. Y., Varma, G., Paus, T., Harris, A. D., Lodygensky, G. A., Beauchamp, M. H., Mabott, D. J., Kopala-Sibley, D. C., Vorstman, J. A. . S., Yeates, K. O., Arnold, P. D., Jacquemont, S., Kadis, D. S., Lippe, S., Bellec, L., Lebel, C., Conrod, P., Bray, S., Wheeler, A. L.

📅 2026-10-31 📖 1 min 📝 222 words BIORXIV NEUROSCIENCE

Summary: Understanding how the human brain matures from infancy through adolescence is essential for identifying factors that influence health outcomes. Magnetic resonance imaging (MRI) provides powerful tools for characterizing structural, microstructural, functional, and metabolic aspects of brain developm...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761379v1?rss=1

Recurrent Spiking Neural Networks for Continuous Motor Control: A Biologically Plausible Model of Motor Cortex

Imamura, H., Imamura, F., Igarashi, J., Isomura, Y., Hira, R.

2026-10-31 1 min 216 words

BIORXIV NEUROSCIENCE

Summary: Motor cortex acquires skilled movements through recurrent spiking dynamics and experience-dependent synaptic plasticity, yet how such circuits learn continuous motor control using biologically local signals remains unclear. Here, we develop a recurrent spiking neural network model of motor cortex th...

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https://www.biorxiv.org/content/10.64898/2026.10.26.756490v1?rss=1

Intelligent clinical decision support system for intracranial aneurysm lesion-based detection using biomedical imaging data

2026-11-01 1 min 0 words

NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-76815-z

Cortex-wide neuronal dynamics underpin quasi-periodic patterns in resting-state fMRI

📅 2026-11-01 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s42003-026-11228-2

Network geometry shapes the transformation of task representations across the human cortex

👤 Michael W. Cole 📅 2026-10-22 📖 1 min 📝 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 22 October 2026; doi:10.1038/s41593-026-02486-8 The dimensionality of intrinsic brain connectivity systematically shapes how task representations compress and expand across the cortex, and a...

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https://www.nature.com/articles/s41593-026-02486-8

Different modes of stimulation at the Neiguan acupoint (PC6) regulate different EEG microstates under VR immersion conditions

Guangjun Wang 2026-10-22 1 min 302 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAcupuncture at the Neiguan acupoint (PC6) alleviates postoperative and chemotherapy-induced nausea and vomiting. Nevertheless, the central regulatory effects of PC6 stimulation on virtual reality motion sickness (VRMS), a form of physiological vertigo triggered by visual-vestibular senso...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1938329

Retraction: Intervention effect of rehabilitation robotic bed under machine learning combined with intensive motor training on stroke patients with hemiplegia

2026-10-14 1 min 0 words FRONTIERS NEUROROBOTICS

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.2023207

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Emerging light-driven neuromorphic hardware for artificial intelligence

👤 Dimitris Tsoukalas 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: Optoelectronic neuromorphic systems have emerged as a promising hardware paradigm for next-generation artificial intelligence, combining the high bandwidth, parallelism, and wavelength selectivity of photonics with the adaptive plasticity of electronic materials. This mini-review provides a focused ...

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Systemic, RPE-directed AAV-tyrosinase therapy restores ocular pigmentation in an OCA1 mouse model

👤 Leah C Byrne 📅 2026-10-31 📖 1 min 📝 71 words LOW VISION

Summary: Oculocutaneous albinism type 1 (OCA1) is a pigmentation disorder caused by biallelic tyrosinase (TYR) mutations. TYR inactivity results in loss of hair, skin, and eye pigment, which is detrimental for ocular function. Hypopigmentation of iris, retinal pigment epithelium (RPE), and choroid results in...

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Systematic review of atropine cost-effectiveness in childhood myopia management

👤 Long Chiau Ming 📅 2026-10-31 📖 1 min 📝 45 words LOW VISION

Summary: CONCLUSION: Studies demonstrate favourable economic value using diverse metrics, including cost per diopter reduction, cost ratios, and cost-utility estimates, rather than uniformly reported ICERs. Greater economic benefits are observed over longer time horizons, largely driven by the avoidance of h...

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Intramedullary Spinal Tuberculosis Mimicking Neoplasm with Ocular Involvement in an Adolescent: A Case Report

👤 Indri Febriani 📅 2026-10-31 📖 1 min 📝 72 words LOW VISION

Summary: Tuberculosis (TB) continues to represent one of the key health problems of high-burden nations. The childhood population is particularly vulnerable to extrapulmonary TB. Spinal intramedullary TB is extremely uncommon and can imitate neoplastic conditions, leading to a late diagnosis with fatal outco...

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Framework-informed analysis of practice: A qualitative study on climate resilience in primary care in Austria

👤 Katharina Brugger 📅 2026-10-31 📖 1 min 📝 59 words LOW VISION

Summary: CONCLUSIONS: PCPs play a vital role in local climate resilience but require tailored, context-specific guidance aligned with PC's integrated, small-scale, multiprofessional nature. Strategic macro-level objectives are required to support low-barrier implementation of resilience measures. Prioritizin...

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Early applications of apple vision pro in clinical medicine and medical education: a scoping review

👤 Chuxu Wang 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: The Apple Vision Pro (AVP), a next-generation spatial computing device, has demonstrated potential applications in multidisciplinary clinical diagnosis, treatment, and medical education, by leveraging its unique technical architecture and performance advantages. In clinical practice, its fundamental...

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A Distributed Vision Transformer-Based Deep Learning Framework for Early Detection and Classification of Lumbar Disc Degeneration

👤 N Chaitanya Kumar
📅 2026-10-31 📖 1 min 📝 68 words

LOW VISION

Summary: Lumbar disc degeneration (LDD) is a principal structural cause of chronic low back pain. Magnetic resonance imaging (MRI) is the reference modality for assessing the intervertebral discs, but severity grading is performed manually, is limited by inter-observer variability, and becomes a bottleneck a...

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A Result-Auditable Protocol for Low-Data Classification of Blue-and-White Porcelain Botanical Motifs Using Frozen Chinese-CLIP Features

👤 Chenghao Yu
📅 2026-10-31 📖 1 min 📝 58 words

LOW VISION

Summary: Low-data visual classification is common in cultural heritage documentation, where expert labels are expensive, and image sources are heterogeneous. This protocol provides result-level audit support for classifying five botanical motifs-peony, lotus, chrysanthemum, plum blossom, and bamboo-on blue-a...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Safety and Efficacy of Anodal transcranial direct-current stimulation over the Dorsolateral Prefrontal Cortex in Post-Stroke Cognitive Impairment: A Systematic Review and Meta-Analysis

👤 Dongyan Wang 📅 2026-10-29 📖 1 min 📝 39 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the DLPFC may improve global cognitive screening performance and ADL in PSCI patients, yet low methodological quality and inadequate safety reporting limit the current evidence. Further high-quality RCTs with rigorous adverse event monitoring are warranted.

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

👤 Zhikun Wu
📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan
📅 2026-10-30 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

Chuhuai Wang
2026-10-31 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

👤 Chen-Hui Pei
📅 2026-10-30 📖 1 min 📝 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Is the surgeon or hospital a larger cost driver for lumbar fusion? A hierarchical Bayesian approach using time-driven activity-based costing

Ahilan Sivaganesan 2026-10-30 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: In cases of single-level lumbar fusions performed at a multihospital health system, variation in cost between surgeons was 30-times greater than variation between hospitals, although residual/unexplained variance was considerable. These hypothesis-generating findings might enhance physi...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

Shouyan Wang 2026-10-30 1 min 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

Aishi Zhou 2026-10-30 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

👤 Jianfeng Hu
📅 2026-10-31 📖 1 min 📝 59 words

BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

Jing Jin 2026-10-31 1 min 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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Monthly Updates [October]

2026-10-01 5 min 1097 words FMHY

Summary: <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   <p>Re-ordered Live Sports based on the <a href="https://challonge.com/ajqvt0wo/fullscreen" rel=...

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https://fmhy.net/posts/oct-2026

ChatGPT and Codex usage limits merging

👤 Zariel 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://help.openai.com/en/articles/20001625-understanding-chatgpt-and-codex-usage-limits Comments URL: <a href="https://news.ycombinator.com/item?id=50320588">https://news....

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Leaking Secrets from Air-Gap Computers by Spelling Radio Signals from RAM (2024)

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Brazilian Court Sanctions Lawyers for Prompt Injection in Court Filing

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Recording and slides now available for ERPLAB Studio webinar

👤 Steve Luck 📅 2024-06-28 📖 1 min 📝 30 words ERP BOOT CAMP

Summary: We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

New Paper: Does the P3b component reflect working memory updating?

👤 Steve Luck 📅 2025-03-21 📖 7 min 📝 1540 words ERP BOOT CAMP

Summary: Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (in press). Enhanced working memory representations for rare events. Psychophysiology. https://doi.org/10.1111/psyp.70038 [<a href="https://doi.org/10.1101/2024.03.20...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/3/20/new-paper-oddball

10-Day ERP Boot Camp to be held June 15-24, 2026 in Davis, California

👤 Steve Luck 📅 2025-08-20 📖 1 min 📝 125 words ERP BOOT CAMP

> **Summary:** We have received another 5 years of funding from the National Institute of Mental Health, so we plan to hold ERP Boot Camps in each of the next 5 summers. The next one will be June 15-24, 2026 in Davis, California. The application portal will open around January 1, 2026. A...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/8/20/boot-camp-summer-2026

Call for Papers: IEEE Brain Special Issue

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 36 words BRAIN

> **Summary:** In a unique interdisciplinary collaboration with the IEEE's Society on Social Implications of Technology (SSIT) and IEEE Brain, J-FLEX is joining forces to explore both the technology of the Internet-of-Medical-Things (IoMT) solutions and medical wearables/ implantables.

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-journal-on-flexible-electronics/

IEEE Brain Joins the American Brain Coalition

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 61 words BRAIN

Summary: IEEE Brain is pleased to announce its acceptance as a nonprofit member of the American Brain Coalition (ABC), a prestigious alliance of over 150 organizations dedicated to advancing brain research, advocacy, and improving treatments for individuals affected by brain conditions. The ABC Board has ent...

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-brain-joins-the-american-brain-coalition-as-a-nonprofit-member/

Call for Papers: IEEE Transactions on Human-Machine Systems

👤 Adriel Carridice 📅 2025-06-18 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/braininsight-articles/call-for-papers-ieee-transactions-on-human-machine-systems/

Q&A with Dr. Alain Dagher, Neurologist and Professor at McGill University and the Montreal Neurological Institute

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

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IEEE Advances Responsible Neurotechnology Standard

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Technical Parameters and Feedback Control for Blood-Brain Barrier Permeability Enhancement by Focused Ultrasound

📅 2025-12-02 📖 1 min 📝 166 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Focused ultrasound combined with intravenously infused microbubbles has been shown to effectively enhance the permeability of the blood-brain barrier, facilitating drug delivery to the brain. A wide range of technical parameters has been evaluated through preclinical studies and clinical trials. Gen...

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Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

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Decoding individual differences in mental information from human brain responses predicted by deep neural networks

📖 1 min 📝 20 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Kiichi Kawahata, Jiaxin Wang, Antoine Blanc, Naoya Maeda, Shinji Nishimoto, Satoshi Nishida

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Beyond rumination: an ecological momentary assessment of daily thinking patterns in maladaptive daydreaming

👤 Lu, A. S., Reich-Fuehrer, M., Andrews, E., Hennessy, S., Christoff Hadjiilieva, K., Kam, J., Andrews-Hanna, J.

📅 2026-10-31 📖 1 min 📝 260 words BIORXIV NEUROSCIENCE

Summary: Maladaptive daydreaming is a form of maladaptive repetitive thought characterized by excessive, dysfunctional fantasy engagement, with notable links to psychological distress and poor well-being. Although symptoms can be reliably measured along a continuum, it remains unclear how they manifest in th...

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Structural Evolution of Neural Networks during Zebrafish Early Development

👤 Wang, Z., Tian, L., Li, B.
📅 2026-10-31 📖 1 min 📝 144 words

BIORXIV NEUROSCIENCE

Summary: The spatial architecture of neural networks underpins nervous system function, yet dynamic in vivo evidence for their early establishment process remains scarce. Here, using long-term confocal live-cell imaging of transgenic zebrafish, we dissected neural network development across the brain, yolk s...

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